YASKAWA

# MOTOPLUS SDK
# FOR VISUAL STUDIO
# USER'S MANUAL

**FOR: DX100, DX200, FS100, YRC1000, AND YRC1000micro**
**YASKAWA ELECTRIC CORPORATION**

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Upon receipt of the product and prior to initial operation, read these instructions thoroughly, and retain for future reference.

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YASKAWA INSTRUCTIONS

MANIPULATOR INSTRUCTIONS
CONTROLLER INSTRUCTIONS
CONTROLLER OPERATOR'S MANUAL
CONTROLLER MAINTENANCE MANUAL

Part Number: 190844-1CD
Revision: 0

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# Safety

## For Your Safety

Robots generally have requirements which are different from other manufacturing equipment, such as larger working areas, high-speed operation, rapid arm movements, etc., which can pose safety hazards.

Read and understand the instruction manuals and related documents, and observe all precautions in order to avoid the risk of injury to personnel and damage to equipment.

Carelessness contributes to serious accidents in the work area.

It is the user's responsibility to ensure that all local, state, and national codes, regulations, rules, or laws relating to safety and safe operating conditions are met and followed.

**DANGER**

- Teaching, operations, and maintenance of the Robot must conform to:
  - Industrial Safety and Health Law
  - Order for Enforcement of the Industrial Safety and Health Law
  - Industrial Safety and Health Regulations
  - Technical Standards for Electrical Facilities

  Other related laws and regulations are:
  - Occupational Safety and Health Act in USA
  - Factory Act (Gewerbeordnung) in Germany
  - Health and Safety at Work, etc. Act in UK
  - EC Machinery Directive 2006/42/EC
- Prepare:
  - SAFETY WORK REGULATIONS
    based on concrete policies for safety management complying with related laws and regulations.
- Observe:
  - JIS B 8433-1: 2015 "Robots for industrial environments-Safety requirements" (ISO 10218-1: 2011) for safe operation of the robot. (JIS B 8433 is for Japan only)
- Reinforce:
  - SAFETY MANAGEMENT SYSTEM
    by designating authorized operators and safety managers for the Robot, as well as giving continuing safety education and training.
- Teaching, operation, and maintenance of the Robot are specified as "Hazardous Operations" in the Industrial Safety and Health Act (for Japan only).
  Personnel engaged in these operations must receive special training offered by YASKAWA.

**DANGER**

- Personnel engaged in operation, maintenance, or management of the Robot must receive required training before using the Robot.

For more information on training, contact a YASKAWA representative.

- Make sure to have and follow all manuals, read them thoroughly and understand the contents of them.

Confirm that you have all required manuals. If any of the manuals are missing, contact a YASKAWA representative.

- Read and understand these instructions throughly before installing, operating, or maintaining the Robot.

Any matter not described in this manual must be regarded as “prohibited” or “improper”.

We suggest that you obtain and review a copy of the ANSI/RIA National Safety Standard for Industrial Robots and Robot Systems (ANSI/RIA R15.06-2012). You can obtain this document from the Robotic Industries Association (RIA) at the following address:

Robotic Industries Association
900 Victors Way
P.O. Box 3724
Ann Arbor, Michigan 48106
TEL: (734) 994-6088
FAX: (734) 994-3338
www.roboticsonline.com

Ultimately, well-trained personnel are the best safeguard against accidents and damage that can result from improper operation of the equipment. The customer is responsible for providing adequately trained personnel to operate, program, and maintain the equipment.

We recommend approved YASKAWA training courses for all personnel involved with the operation, programming, or repair of the equipment.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

## WARNING

- Safe operation of this equipment is the users responsibility.
  - The conditions under which the equipment will be operated safely should be reviewed by the user. The user must be aware of the various national codes, ANSI/RIA R15.06-2012 safety standards, and other local codes that may pertain to the installation and use of this equipment.

Not following all national codes, safety standards and local codes can result in death or serious injury

- Additional safety measures for personnel and equipment may be required depending on system installation, operation, and/or location.
  - The following safety equipment is provided as standard:
    - Safety barriers
    - Door interlocks
    - EMERGENCY STOP button

Not providing additional safety measures as required can result in death or serious injury.

- Check all safety equipment frequently for proper operation.
  - Repair or replace any non-functioning safety equipment immediately.

If safety equipment does not operate properly death or serious injury can result.

## CAUTION

- Only trained personnel familiar with the operation, manuals, electrical design, and interconnections of this equipment should program, or maintain the system.

Any personnel involved with the operation of the equipment must understand potential dangers of operation.

## NOTICE

- The drawings and photos in this manual are examples. Differences may exist between them and the delivered product.
- YASKAWA may modify this model without notice due to product improvements, modifications, or changes in specifications. If such modification is made, the manual number will also be revised.
- Some operations require standard passwords and while others require special passwords.
- If a manual is damaged or lost, contact a YASKAWA representative to order a new copy. Make sure to tell the YASKAWA representative the Part Number listed on the front cover.

## Notes for Safe Operation

Read this manual carefully before installing, operating, maintaining, or inspecting the system.

In this instruction, Safe Operations are classified as “DANGER”, “WARNING”, “CAUTION” or “NOTICE”.

| | |
|---|---|
| DANGER | Indicates an imminently hazardous situation which, if not avoided, **WILL result in death or serious injury**. |
| WARNING | Indicates a potentially hazardous situation which, if not avoided, **MAY result in death or serious injury**. |
| CAUTION | Indicates a hazardous situation, which if not avoided, **MAY result in minor to moderate injury**. |
| CAUTION | Indicate a situation which if not avoided **may result in equipment damage.** |
| NOTICE | Indicates **practices not related to personal injury**. |

| NOTICE |
|---|
| To ensure safe and efficient operation at all times, be sure to follow all instructions, even if not designated as “DANGER”, “WARNING” or “CAUTION”. |

## Installation and Wiring Safety

Review the Manipulator and Controller Instructions for details on installation and wiring.

In planning installation, adapt an easy to observe arrangement to ensure safety. Take safety into consideration when planning the installation. Observe the following when installing the Robot:

**DANGER**

- Maintenance and inspection must be performed by specified personnel.

Failure to observe this caution may result in electric shock or injury.

- For disassembly or repair, contact a YASKAWA representative.
- Do not remove the motor, and do not release the brake.

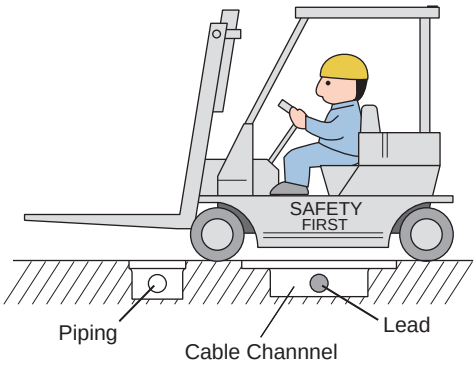
Failure to observe these safety precautions may result in death or serious injury from unexpected motion of the Robot's arm.

- Any person who programs, teaches, operates, maintains or repairs the included system MUST be trained and demonstrates competence to safely perform assigned tasks.

Failure to observe these safety precautions may result in death or serious injury from unexpected movements.

**WARNING**

- Run the piping, wiring, and cables for the Controller, Robot, Positioner control panel, peripheral devices, etc. in a pit so that are not stepped on by personnel or run over by a forklift.




Failure to observe this Warning may cause personnel to trip over exposed piping, wiring, or a cables, which may result in personal injury. Additionally it may also cause damage to piping, wiring, or cables, and unexpected movement of the Robot, which may result in personal injury and/or equipment damage.

## CAUTION

- Make sure all covers and shields are installed correctly before operating.
  - Some drawings in this manual may have protective covers or shields removed to show details.

Not having all covers and shields installed correctly can result in injury.

- When installing the Robot system, avoid interference with buildings, structures, utilities, other machines.

Not avoiding these items may create trapping or pinch points.

- Do not make unauthorized modifications.

Unauthorized modifications can result in injury or equipment damage and will void the warranty.

- Inspect:
  - For problems with movement
  - Damages to external wires

Repair any problems immediately and perform all necessary procedures. If problems are not repaired or procedures are not fixed unexpected results can occur causing injury.

## NOTICE

- If supplying a supplementary audible means for Robot operation, it shall exceed the ambient noise at the end-use of the application.
- Any changes or additions to the applicable information as provided by the manufacturer is to be provided by the party that makes the change or addition to the Robot system

## Ensure Safety

**DANGER**

- When the power supplies of the Robot are turned ON at start-up, be sure to confirm the following:
  - Safety protection devices such as the E-STOP circuit, door interlocks, etc. operate normally.
  - Each axis operates normally in TEACH mode.
  - Robot operates normally at the speed limit or less in the TEACH mode.(Speed limit: 250 mm/s at the TCP or the flange)
  - The teaching function and the playback function operate normally.
- The Robot may stop movements while waiting for a condition to be satisfied during operation.

Once meeting the condition, the Robot starts movement causing a danger that will cause death or severe injury.

- Make sure to clearly indicate when the Robot is in operation:
  - Use a pilot lamp and/or an audible alert or
  - The Robot stops operation if the operator comes close.
- Install a safety fence around the Robot to prevent any accidental contact with the Robot when power is applied.
  - Display a warning sign stating “Off-Limits During Operation” at the entrance of the safety fence.
  - The gate of the safety fence must be equipped with a safety interlock to turn the servo power OFF when the gate opens.
  - Make sure interlocks operate properly before use.
- For areas not enclosed by safety fences, use a photoelectric sensor, a safety light curtain, etc. to make sure that the Robot stops its operation if the operator enters its operating range.

Failure to observe this DANGER will result in death or serious injury due to contact with the Robot.

All personnel working with the Robot (safety administration, installation, operation, and maintenance personnel) must always be prepared and “Safety First” minded, to ensure the safety of all personnel.

**WARNING**

- In the vicinity of the area where the Robot is installed, avoid any dangerous actions, such as entering the Robot's operating range without due care.

Failure to observe this instruction may cause contact with the Robot or peripheral equipment, which may result in death or serious injury.

- Strictly observe the safety precautions and signs in the factory, such as “Flammable”, “High Voltage”, “Danger”, “Off-limits to Unauthorized Personnel”.

Failure to observe this instruction may result in death or serious injury do to fire, electric shock, caused by contact with the Robot or other equipment.

- Strictly observe the following precautions about clothing:
  - Always wear approved work clothes (no loose-fitting clothes).
  - To prevent mis-operation, do not wear gloves when operating the Robot.
  - Do not allow underwear, shirts, or neckties hang out from the work clothes.
  - Do not wear accessories, such as earrings, rings, or necklaces.
  - Always wear protective safety equipment, such as hard hats, safety shoes (with slip-proof soles), face shields, safety glasses, and gloves as necessary.

Failure to observe this instruction may result in death or serious injury.

- The following must be understood and strictly observed by all personnel as rules:
  - Unauthorized personnel other than the operator must not approach the area where the Robot is installed.

Failure to observe this instruction may cause contact with the Robot, Controller, control panel, workpiece, or Positioner, etc., may result in death or serious injury.

## DX100, DX200, and YRC1000 Controllers

**WARNING**

- Turn OFF servo power before operating.
  - Press the EMERGENCY STOP button to turn off SERVO POWER. When servo power is OFF, the SERVO ON LED on the Programming Pendant is OFF.

If the EMERGENCY STOP button(s) do not work correctly, death or serious injury may result. Do not use if the EMERGENCY STOP button does not perform correctly.

*Fig. : EMERGENCY STOP Button*

- Clear the cell of all items which could interfere with the operation before releasing the EMERGENCY STOP button.

Death or serious injury may result from unintentional or unexpected motion.

*Fig. : Release of EMERGENCY STOP Button*

- Make sure no person is in the operating range and the operator is in a safe location before:
  - Turning ON power to the Controller
  - Moving the Robot with the Programming Pendant
  - Running the system in the TEACH mode
  - Performing automatic operations

Death or serious injury may result if a person enters the operating range during operation. Immediately press an EMERGENCY STOP button whenever there is a problem.

## FS100 and YRC1000micro Controller

**WARNING**

- Before operating the manipulator, check that servo power is turned OFF when the emergency stop button on the programming pendant is pressed.
  When the servo power is turned OFF, the SERVO ON LED on the programing pendant is turned OFF.

Injury or damage to machinery may result if the emergency stop circuit cannot stop the manipulator during an emergency. The manipulator should not be used if the emergency stop button does not function.

*Figure 1: Emergency Stop Button*

- In the case of not using the programming pendant, be sure to supply the emergency stop button on the equipment. Then before operating the manipulator, check to be sure that the servo power is turned OFF by pressing the emergency stop button.
  Connect the external emergency stop button to the 5-6 pin and 16-17 pin of the robot system signal connector (CN2).

- Upon shipment of the robot controller, this signal is connected by a safety signal short circuit connector. To use the signal, make sure to supply a new connector, and then input it.

If the signal is input with the safety signal short circuit connector, it does not function, which may result in personal injury or equipment damage.

- Once the emergency stop button is released, clear the cell of all items which could interfere with the operation of the manipulator. Then turn the servo power ON.

Injury may result from unintentional or unexpected manipulator motion.

*Figure 2: Release of Emergency Stop Button*

- Observe the following precautions when performing teaching operations within the manipulator's operating range:
  - View the manipulator from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Ensure that you have a safe place to retreat in case of emergency.

Improper or unintended manipulator operation may result in injury.

**CAUTION**

- All operators, programmers, maintenance personnel, supervisors, and anyone working near the system must be familiar with the operation of this equipment.
  - All personnel involved with the operation of the equipment must understand potential dangers of operation.
- General safeguarding tips:
  - Place system in E-STOP mode whenever it is not in use.
  - Use lockout/tagout procedures during equipment maintenance in accordance with ANSI/RIA R15.06-2012, section 4.2.5, Sources of Energy. Refer also to Section 1910.147 (29CFR, Part 1910), Occupational Safety and Health Standards for General Industry (OSHA).
  - Only trained personnel familiar with the operation of this equipment, the operator's manuals, the system equipment, and options and accessories can operate equipment.

Improper operation can result in personal injury and/or damage to the equipment.

- If the light in the operator's working space is not bright enough, provide the space with appropriate lighting.

**CAUTION**

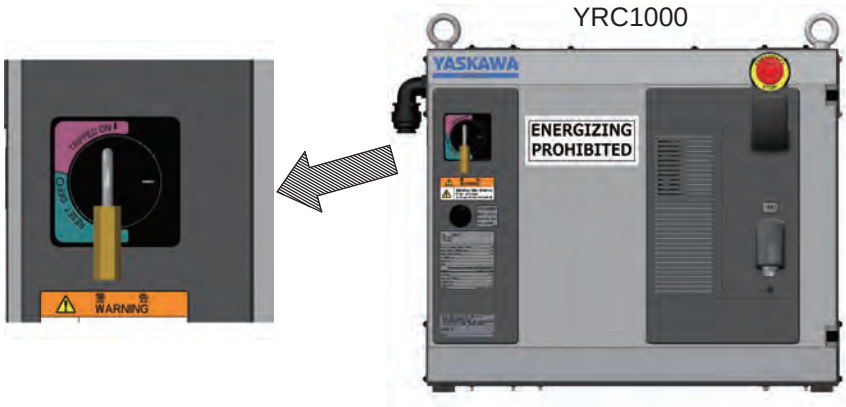
- Store industrial tools, etc. in a safe location outside the Robot's operating range.

If an industrial tool, etc. is left unattended on the Robot, on a fixture, or on the floor, etc., the Robot may come in contact with the industrial tool left unattended, which may result in damage to the Robot and/or the fixture.

## Operation Safety

**DANGER**

- Personnel engaged in teaching or inspection, etc. of the Robot must receive special training required by applicable laws and regulations.
- While performing inspection and maintenance, wiring, or attaching a tool to the Robot, etc., make sure to turn OFF the power supply of the Robot Controller and the tool, and keep the switch of the power supply locked so that unauthorized personnel cannot turn ON the power supply.
  In addition, display a warning sign stating “Energizing Prohibited”.




  Turning ON the power supply without due care during inspection and maintenance, etc., may cause electric shock or unexpected movement of the Robot, which may result in personal injury.

- Use the Robot only within the specifications described in the manuals for the Robot.

Failure to observe this instruction may result in personal injury and/or equipment damage.

- Observe the following precautions when performing a teaching operation within the Robot's operating range:
  - Be sure to perform lockout by putting a lockout device on the safety fence when going into the area enclosed by the safety fence. In addition, the operator of the teaching operation must display the sign that the operation is being performed so that no other person closes the safety fence.
  - View the Robot from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Always keep in mind emergency response measures against the Robot’s unexpected movement toward a person.
  - Ensure a safe place to retreat in case of emergency.

Failure to observe this instruction may cause improper or unintended movement of the Robot, which may result in personal injury.

**DANGER**

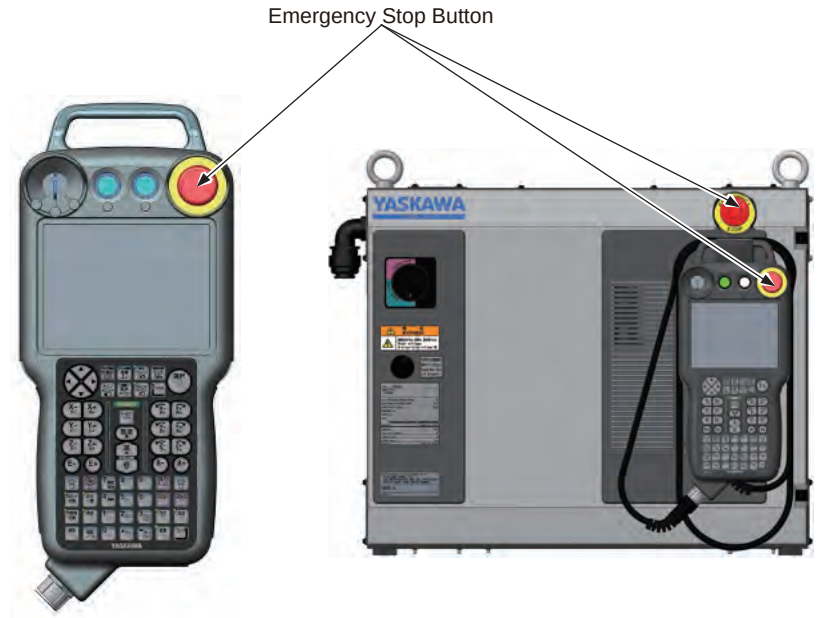
- Before operating the Robot, make sure the servo power is turned OFF by performing the following operations. When the servo power is turned OFF, the SERVO ON LED on the Programming Pendant is turned OFF.
  - Press the EMERGENCY STOP buttons on the front door of the Robot Controller, on the Programming Pendant, on the external control device, etc.
  - Disconnect the safety plug of the safety fence. (when in the PLAY mode or REMOTE mode)

If operation of the Robot cannot be stopped in an emergency, personal injury and/or equipment damage may result.

- Make sure that all safety protection devices are activated before starting a job in the PLAY mode.
- Confirm that no person is present in the Robot's operating range and that the operator is in a safe location before:
  - Turning ON the Robot Controller
  - Moving the Robot by using the Programming Pendant
  - Running the system in the TEACH mode
  - Performing automatic operations

Personal injury may result if a person enters the Robot's operating range during operation

- Immediately press an EMERGENCY STOP button whenever there is a problem.




Programming Pendant

YRC1000

**WARNING**

- Read “Safety” of the Robot Controller instructions before operating.

Not reading and understanding chapter 1 of the Robot Controller instruction can result in death or serious injury.

- Read and understand all Warning Labels before operating.

Not reading and understanding all Warning Labels can result in death or serious injury.

- Confirm that no person is present in the P-point maximum envelope of the Robot before:
  - Turning on the power for the Robot Controller.
  - Moving the Robot with the Programming Pendant.
  - Running the system in the TEACH mode.
  - Performing automatic operations.

Injury may result if anyone enters the working envelope of the Robot during operation. Always press an EMERGENCY STOP button immediately if there are problems.

- Observe the following when performing teaching operation within the operating range:
  - Lockout by putting a lockout device on the safety fence when going into the area enclosed by the safety fence.
  - Display a sign that operations are being performed so no other person closes the safety fence.
  - View from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Always keep in mind emergency response measures against unexpected movement toward a person.
  - Ensure a safe place to retreat in case of emergency.

Failure to observe this precautions may cause improper or unintended movement, which may result in personal injury.

- Maintenance and inspection must be performed by specified personnel.

Failure to observe this Warning may result in electric shock or injury.

- Contact a YASKAWA representative for disassembly or repairs.

Not contacting a YASKAWA representative can result in electrical shock or injury.

## CAUTION

Robot Cells have Collaborative Motion functionality:

Collaboration is a special type of operation between a person and Robot sharing a common workspace. The following are the guidelines for collaborative operation.

1. Used for pre-determined tasks.
2. Possible when all protective measures are active.
3. For Robots with features specifically designed for collaborative operation.

The integrator shall include in the information for use the safeguards and mode selection required for collaborative operation.

## CAUTION

- Do not operate the Robot when a [COOLING FAN2 ERROR] appears on the Programming Pendant.

If operation continues with a warning message, equipment damage can occur.

- During high speed continuous operation Robot temperature may rise quickly depending on ambient temperature and operation pattern.

If a warning message displays stop operations or equipment damage may occur.

- Monitor warning messages on the Programming Pendant.

Not monitoring warning messages may cause equipment damage.

- Refer to the Controller Concurrent I/O manual for details on the signal output.

Not referring to Controller Concurrent I/O manual can result in equipment damage.

## Maintenance Safety

**WARNING**

- Make sure equipment has no potentially hazardous conditions.
  - area is clean and free of water, oil, debris, etc.
  - all safeguards are in place.
  - all safety equipment work correctly. Repair or replace any non-functioning safety equipment immediately.
  - Check the EMEREGENCY STOP button(s) for proper operation before programming. The equipment must be in E-STOP mode when not in use.

If a hazardous condition is present death or serious injury may occur.

- Use care when modifying software.
  - The equipment allows modifications to the software for maximum performance.

All modifications made to the software will change the way the equipment operates and may cause death or serious injury, as well as damage parts of the system.

- Make sure all modifications did not make create a hazardous or dangerous condition in all modes.

All modifications made to the software will change the way the equipment operates and may cause death or serious injury, as well as damage parts of the system.

- Disconnect and lockout/tagout all sources of energy before making modifications or connections.

Not disconnecting and doing lockout/tagout of all sources of energy can result in death or serious injury.

- Read and understand all maintenance procedures before completing procedures.

Not reading and understanding maintenance procedure may result in death or serious injury.

**CAUTION**

- Do not modify the Robot Controller.

Making modifications without written permission from YASKAWA will void the warranty.

- Back up all programs and jobs onto suitable media before program changes are made.

To avoid loss of information, programs, or jobs, a backup must always be made before any service procedures are done and before any changes are made to options, accessories, or equipment.

- Use proper replacement parts only.

Not using proper replacement parts can cause damage to equipment.

- All connections must be made within the standard voltage and current ratings of the equipment.

Improper connections can damage the equipment.

## Notes for Moving and Transferring the Robot

**DANGER**

- When relocating, transferring, or selling the Robot, make sure that the Robot is always accompanied by its manuals so that all users have access to necessary manuals.

See the Bill of Material for a list of the manuals.

If any of them is missing, contact a YASKAWA representative. The telephone numbers of our offices are listed on the back cover of this manual.

- If a warning label on the Manipulator or the Controller is dirty and unreadable, clean the label to make it clearly readable. If a warning label has come off, put the label back in place. Note that some local laws and regulations may prohibit equipment operation if safety labels are not in place.

Contact a YASKAWA representative if requiring new warning labels.

- After the Robot is relocated, inspection by a YASKAWA representative is recommended.

If installation or wiring of a device is incorrect, personal injury and/or equipment damage may result.

## Definition of Terms Used Often in This Manual

The Robot is the YASKAWA industrial robot product.

The Robot usually consists of a Manipulator, Robot Controller, Programming Pendant, and Robot cables.

In this manual, the equipment is designated as follows:

| Equipment | Manual Designation |
|---|---|
| DX100, DX200, FS100, YRC1000 and YRC1000micro controller | Robot Controller |
| DX100, DX200, FS100, YRC1000 and YRC1000micro Programming Pendant | Programming Pendant |
| Manipulator | Manipulator |
| Cable between the Robot and the Controller | Robot cable |
| Positioner | Positioner |
| Controller, Programming Pendant, Manipulator | Robot |
| Robot Controller (FS100 and YRC1000micro) Programming Pendant safety signal short circuit connector | safety signal short circuit connector |

Descriptions of the Programming Pendant keys, buttons, and displays are shown as follows:

| Equipment | | Manual Designation |
|---|---|---|
| Programming Pendant | Character Keys /Symbol Keys | The keys which have characters or symbols printed on them are denoted with [ ].<br>e.g. [ENTER] |
| | Axis Keys /Numeric Keys | [Axis Key] and [Numeric Key] are generic names for the keys for axis operation and number input. |
| | Keys pressed simultaneously | When two keys are to be pressed simultaneously, the keys are shown with a “+” sign between them, e.g. [SHIFT]+[COORD]. |
| | Mode Switch | Mode Switch can select three kinds of modes that are denoted as follows: REMOTE, PLAY or TEACH.<br>(The switch names are denoted as symbols) |
| | Button | The three buttons on the upper side of the Programming Pendant are denoted as follows: START, HOLD, or EMERGENCY STOP.<br>(The button names are denoted as symbols) |
| | Displays | The menu displayed in the Programming Pendant is denoted with { }.<br>e.g. {JOB} |

## Registered Trademark

In this manual, names of companies, corporations, or products are trademarks, registered trademarks, or bland names for each company or corporation. The indications of ® and ™ are omitted.

# Robot Disposal

**WARNING**

- Take precautionary measures to prevent the Robot from overturning, such as anchoring it firmly, etc., even when temporarily storing it before disposal.

Failure to observe this instruction may cause overturning of the Robot, which may result in personal injury.

**CAUTION**

- Do not modify the Robot

Failure to observe this instruction can cause fire, mechanical failure, or malfunction, which may result in personal injury and/or equipment damage.

**NOTICE**

- When disposing of or recycling the Robot, follow the applicable national/local laws and regulations.
- This symbol is applicable in some locations.

The wheelie bin symbol on this product, manual or its packaging indicates that at the end of life the product should enter the recycling system. It must be disposed at an appropriate collection point for electrical and electronic equipment (EEE) and should not be put in the normal waste stream.

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# 1 Outline

## 1.1 Introduction

MotoPlus Software Development Kit (SDK) is used to develop customer specific applications that operate on the main CPU board of the YASKAWA robot controller. The application is run concurrently to the robot firmware, in real time. Programming of the application is done using C language with Microsoft Visual Studio. Users can develop a variety of application programs on their own without modifying the source code of the robot controller.

This advanced toolset is intended only for experienced software developers who are familiar with the C programming language.

## 1.2 What MotoPlus Can Do

Users can use the C language, a general programming language, and MotoPlus libraries to develop a customer specific application program in which various types of resources are used, e.g., the Ethernet communication port, the RS232C serial communication port, and the programming pendant of the robot controller.

### 1.2.1 Supported Services

MotoPlus SDK provides users with a variety of services as follows:

- **Application task control**
  Application task (thread) start/stop, data transmission between application programs by the mailbox and semaphore, exclusive control and execution synchronization

- **Robot control**
  Robot control from the application program

- **JOB control**
  Start-stop control of the JOB by the application, data transmission between Jobs by variables (byte, integer, double-precision, floating point, character string, and position variable), and execution synchronization

- **CIO control**
  I/O between the application program and the robot CIO including read/write access to registers

- **Ethernet communication control**
  Industry standard Ethernet (TCP/IP) communication available through standard sockets

- **Programming pendant communication**
  Data transmission between the application and the programming pendant application

- **Robot event notification**
  Event notification to the application at every I/O control cycle and interpolation control cycle of the system

- **RS232C serial communication control**
  RS232C serial communication from the application

- **Sensor control**
  Robot path correction, adjustment, and control can be achieved while reacting to external sensors

- **Memory management**
  Dynamic and static memory allocation

- **General-purpose file control (Not Available on DX100)**
  Interface with custom data files to save/load application settings

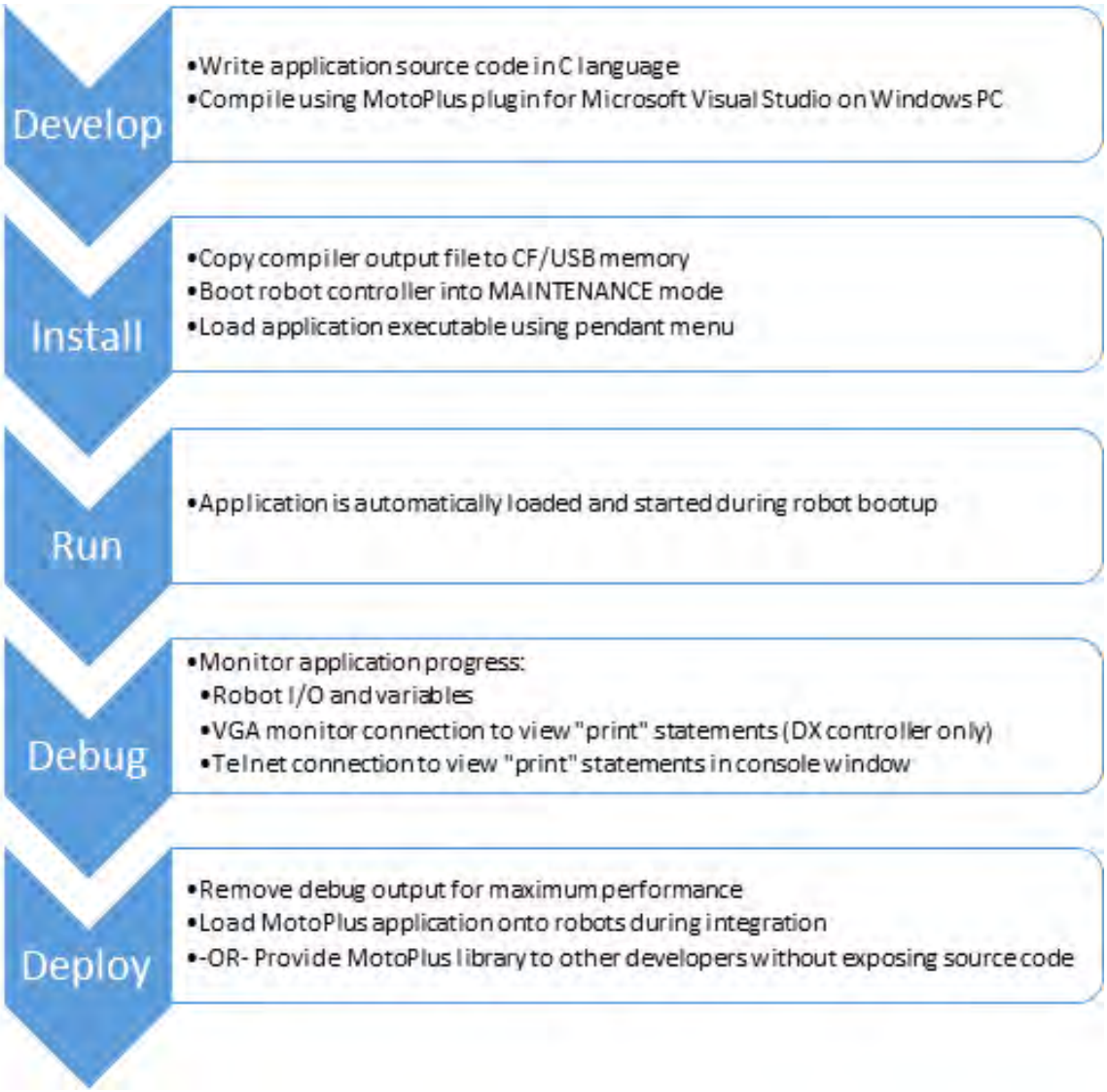
- **Existing file control (Not Available on DX100)**
  Interface with standard robot files (DAT, CND, JBI, etc…)

- **Servo control**
  Control servo power for automated robot operation

- **User watchdog**
  Watchdog to monitor whether the application operates normally

- **Coordinate conversion (Not Available on DX100)**
  Forward and inverse kinematic calculations available for position and frame calculations

## 1.3 Development Process



## 1.4 Notices

**NOTICE**

- C language allows for dynamic memory operations. When programming in C language, pointers with memory addresses may be used. If the pointer variable is set incorrectly, the system memory area may be rewritten.
  If the system memory area is rewritten, critical problems occur, e.g., the software hangs up (the hang-up status is detected by a CPU exception or the watchdog check function, then the servo power turns OFF and the system shuts down), the robot stops its operation due to an alarm, or the programming pendant becomes inoperable. If the system memory operation is incorrect, the application itself may not operate as it is designed.
- The task “mpUsrRoot”, which starts when the power is turned ON, has a high priority to start other application tasks and initialize the entire application quickly. Due to its high task priority, if a time-consuming process (0.1 milliseconds or longer) is done in this task, the processing time for the robot control becomes insufficient. Then system alarms may occur or the programming pendant or the HOLD button may freeze. Thus, as shown in the sample program (refer to Task Control Sample Program” in “Programmer’s Manual for New Language Environment MotoPlus”, make sure to complete the task “mpUsrRoot” after starting up another application or creating a semaphore.
- Debugging via Telnet

1. After completing debugging, set the Telnet function to INVALID (S2C1119 = 0). This is because the Telnet function consumes system processing time, and an alarm may occur due to insufficient processing time for robot operation.
2. With the Telnet function, debug only tasks with normal priority. For debugging the tasks with high priority (the tasks started with the task priority: MP_PRI_IO_CLK_TAKE, MP_PRI_IP_CLK_TAKE, or MP_TIME_CRITICAL, or task “mpUsrRoot”), use the robot variables visible on the programming pendant. This is because, if printf() or puts() is done with a high priority task, processing time for robot operation control is consumed, and a system alarm may occur due to insufficient processing time.
3. After completing debugging, make sure to disable the debugging code using printf() or puts(). This is because the processing time of printf() and puts() influences the system, and an error may occur such as insufficient processing time during robot operation.

When using VxWorks SHELL commands, which become available by connecting with the controller via Telnet, a system alarm may occur due to insufficient processing time for robot operation. Do NOT use the VxWorks SHELL commands.

# 2 SDK Installation and Setup

## 2.1 System Requirements

The following requirements must be met for the host computer used to develop the MotoPlus application:

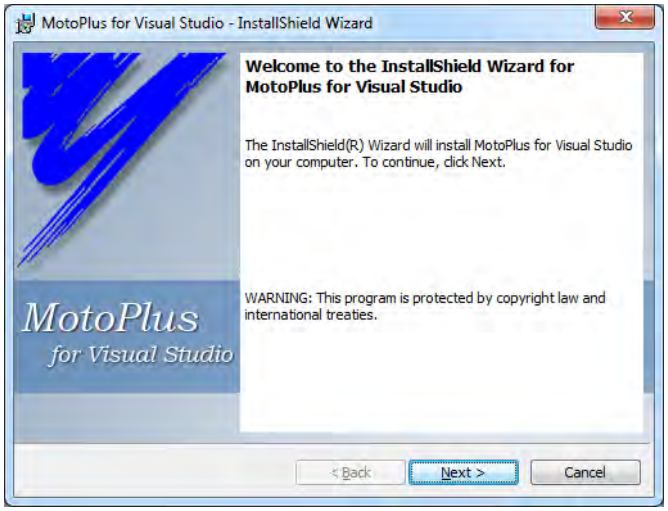
| | |
|---|---|
| PC with Microsoft Windows | Windows XP SP2, Vista, 7, 8, or 10 |
| .Net Framework | 3.5 |
| USB Hardware Key Dongle | Included with SDK package |
| Microsoft Visual Studio | 2008 Standard, Professional<br>2010 Professional, Premium, Ultimate<br>Visual C++ 2010 Express (free download from Microsoft)<br>2012 Professional, Premium, Ultimate<br>2012 Express for Windows Desktop (free download from Microsoft)<br>2013 Professional, Premium, Ultimate<br>2013 Express for Windows Desktop (free download from Microsoft)<br>2015 Professional, Enterprise<br>2015 Community (free download from Microsoft)<br>2017 Professional, Enterprise<br>2017 Community (free download from Microsoft)<br><br>*Not compatible with “Visual Studio Code”.* |
| External memory card to load on robot pendant | Compact Flash (CF) or USB Flash (USB) |
| User Rights on PC | **Administrator rights required** |

## 2.2 Installation of SDK

**NOTICE**

Do not insert the USB hardware key dongle into the computer until after installing the Sentinel driver.

1. Insert the MotoPlus SDK for Visual Studio disc into your Windows PC. If AutoRun is enabled, the installation will automatically start. Otherwise, run “setup.exe” from the root of the DVD.




**NOTICE**

The license agreement is located on the root of your installation DVD. Please read this prior to installing the software. You will be required to agree to the terms of the licensing.

2. Click [Next] and continue through the installation.
3. When installation is complete, you must reboot the computer to register system environment variables required by the compiler.
4. After restarting the PC, please install the Sentinel driver located on the root of your installation DVD. This is required for the USB hardware key dongle.

**NOTICE**

If you are installing the Sentinel driver on a notebook/laptop computer, it is recommended to first remove the computer from any docking station(s) prior to installation.

5. Insert the USB hardware key dongle into your PC.

Installation of the SDK is now complete.

## 2.3 Setup of Robot Controller

### 2.3.1 Install VGA Adapter Cable for Debugging (Not Available on FS100 or YRC1000micro)

The SDK package comes with a VGA adapter cable. This cable allows you to connect a standard VGA computer monitor to view program output. It should only be used for application debugging purposes and not shipped with a production robot controller.

**CAUTION**

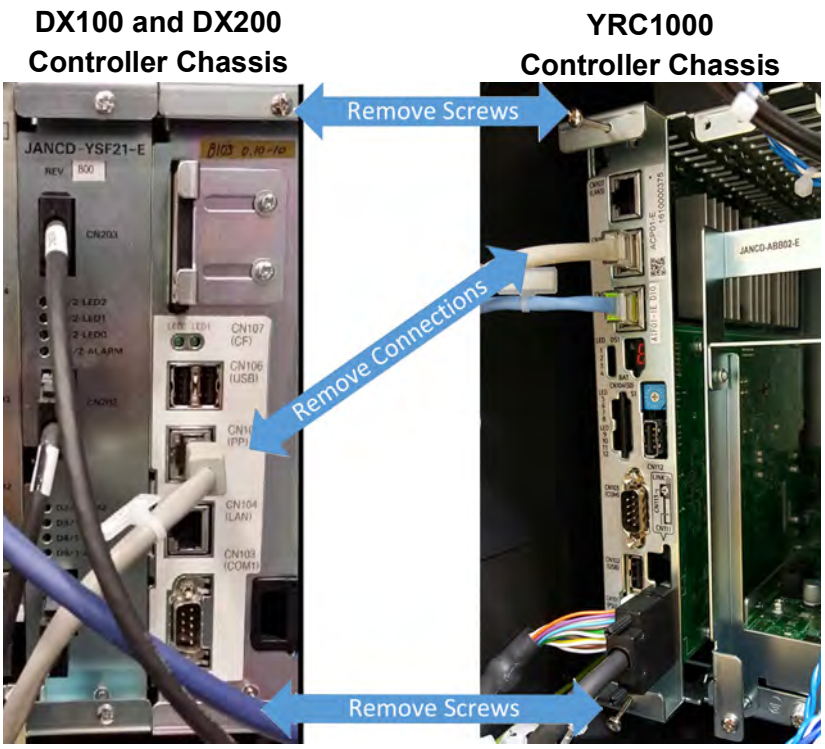
- Follow standard ESD safe procedures when handling the CPU board!

Static electricity can severely damage sensitive electronics.

1. Power off the robot controller.
2. Locate and remove the CPU board in the controller chassis. See Table for board number. The board is secured with two screws on top and bottom. You may also have to temporarily disconnect all cables going to connectors CN103, CN104, CN105 and CN106.

*Table 2-1: CPU Board in Controller Chassis.*

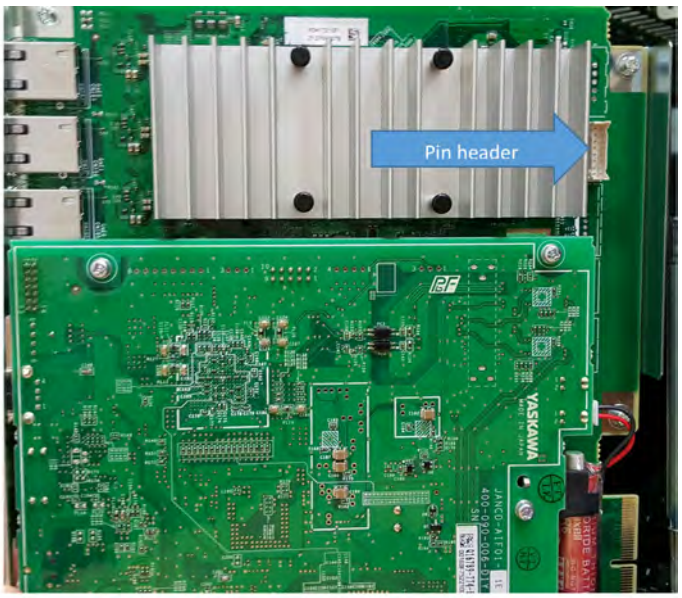
| Controller | CPU Board Part Number |
|---|---|
| DX100 | JANCD-YCP01 |
| DX200 | JANCD-YCP21 |
| YRC1000 | ACP01 |



3. Connect the VGA adapter cable to the pin header on the CPU board and route the cable out of the chassis. If equipped carefully open the plastic tab which is securing the opening and install the cable.

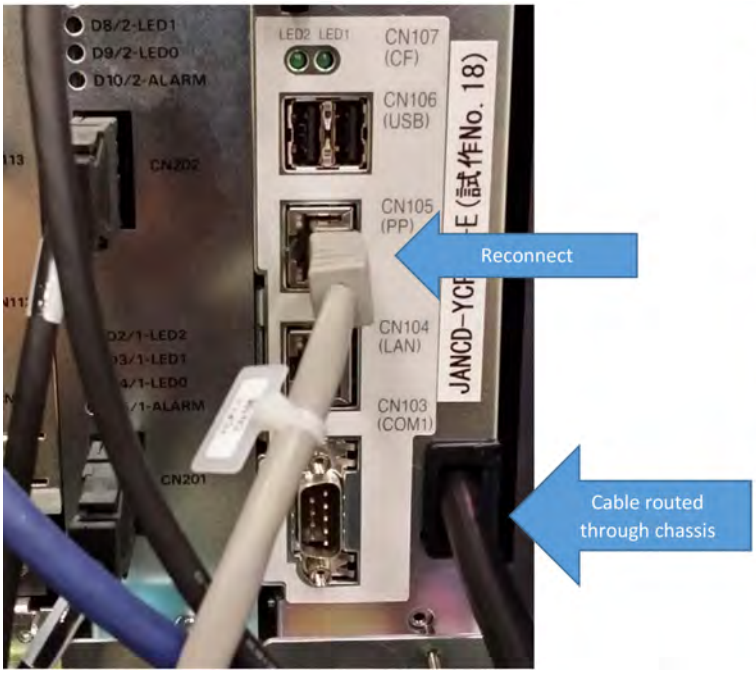
**DX100 and DX200**
**Pin Header**

**YRC1000**
**Pin Header**

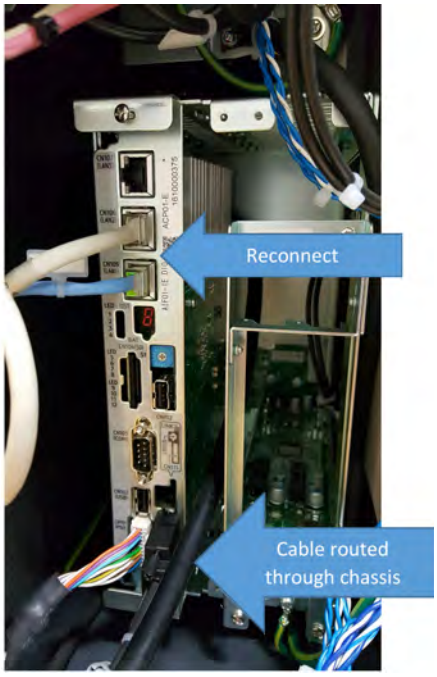



4. Reinsert the CPU board into the controller chassis, tighten the screws securing the board, and reconnect all cables removed in the previous steps.

**DX100 and DX200**
**Reassembly**




**YRC1000**
**Reassembly**

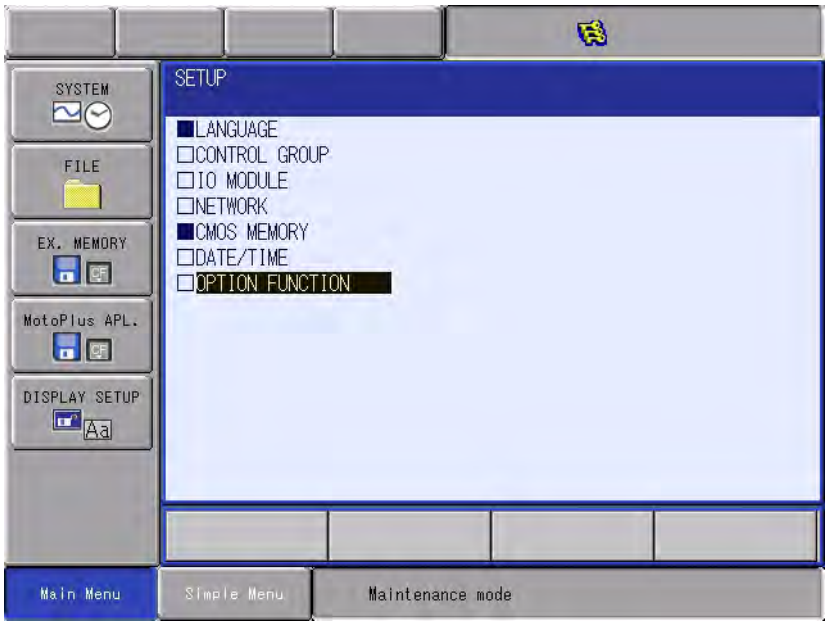


### 2.3.2 Enabling MotoPlus (Not Available on DX100)

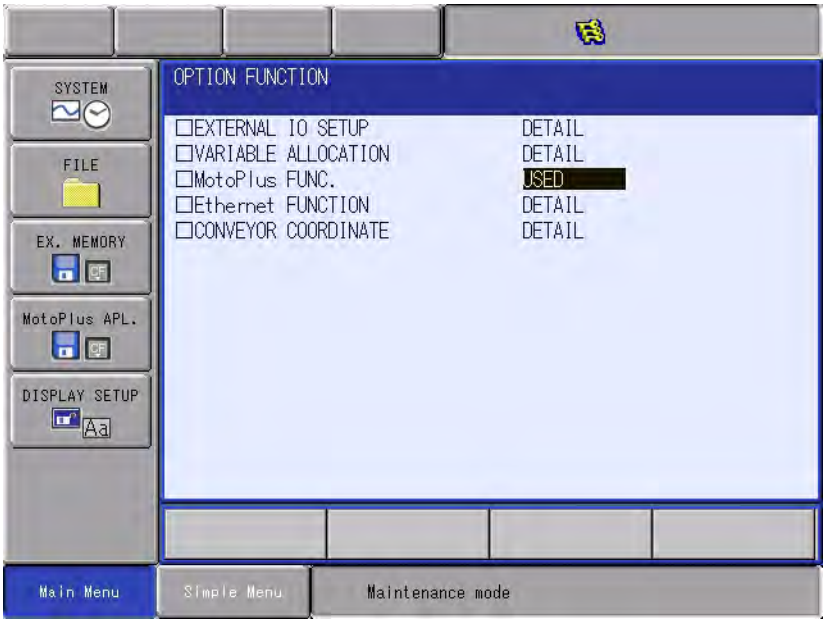
The MotoPlus function must be enabled in MAINTENANCE mode.

1. Boot the robot controller in MAINTENANCE mode.
2. Upgrade to MANAGEMENT level security. Touch [System] > [Setup].

3. Select “OPTION FUNCTION”.



4. Set “MotoPlus FUNC.” to “USED”. Press {ENTER} to save your change. If you are asked to “Initialize related file SRAM.DAT”, touch *[YES]*.




5. *Set “MOTOMAN DRIVER” to “USED”.*
   - Make sure to set this AFTER “MotoPlus FUNC.” If these are set in the wrong order, please set both functions to “NOT USED” and then toggle them back to “USED” in the correct order.

**NOTICE**

The MOTOMAN DRIVER option is only available if the controller has the following software version or greater:

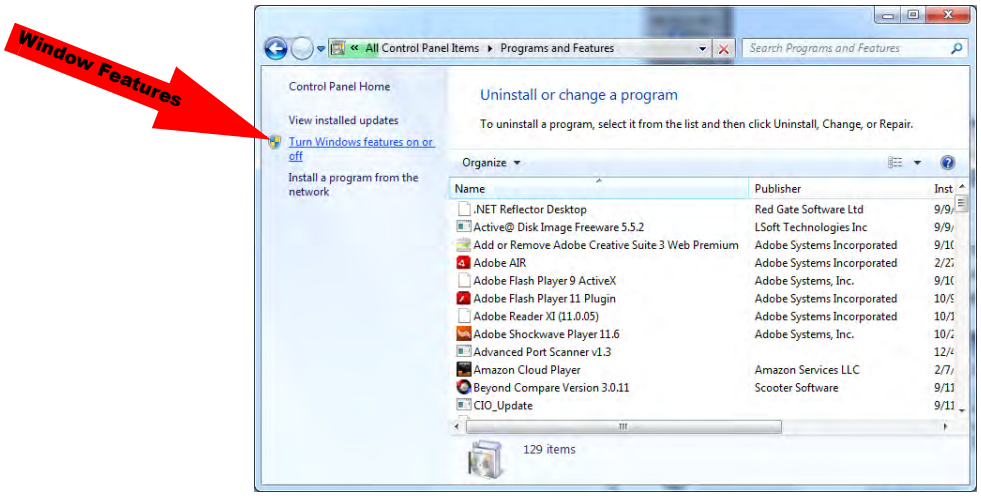
- FS100: FS3.30.00-00
- DX200: DN2.21.00-00
- YRC1000: YAS1.11.00-00
- YRC1000micro: Any

## 2.4 Enable Telnet for Debugging

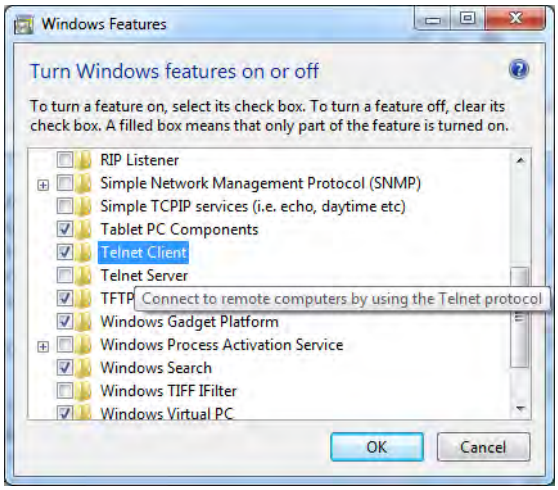
One of the options for debugging is to connect to the robot using a telnet connection. (See *Chapter 5 "Debugging"*).

On Windows 7, the default telnet console is not enabled by default. This must be enabled prior to debugging.

1. Click the Windows START orb, and type "programs and features", then press [Enter].
2. From the menu on the left, select "Turn Windows features on or off".




3. In the Windows Features list, check the box next to "Telnet Client" and click [OK].



# 3 Creating MotoPlus Application

## 3.1 Rules for Application Source Code

Be sure to observe the following rules when creating source code.

Refer to the MotoPlus Programmer's manual for detailed information on programming configuration and sample programs.

**NOTICE**

Sample source code files are located in your installation directory and on the MotoPlus for Visual Studio installation disc.

Please remember that the source code should follow ANSI-C syntax. The only exception is that users should not define a “main (...)” function. This is replaced by “mpUsrRoot(...)”.

1. Up to 32 one-byte characters can be used for a file name. Up to 16 two-byte characters can also be used. File names are case-sensitive. Set '.c' as the file extension. (C++ library is not fully supported on the MotoPlus platform)
2. Put “#include <MotoPlus.h>” at the top of each source code file. When “MotoPlus.h” is included, all services provided by MotoPlus are available.
3. MotoPlus applications do not have a “main” function. This is replaced by “mpUsrRoot”. This function must have the following signature:

   void mpUsrRoot (int arg1, int arg2, int arg3, int arg4, int arg5, int arg6, int arg7, int arg8, int arg9, int arg10)

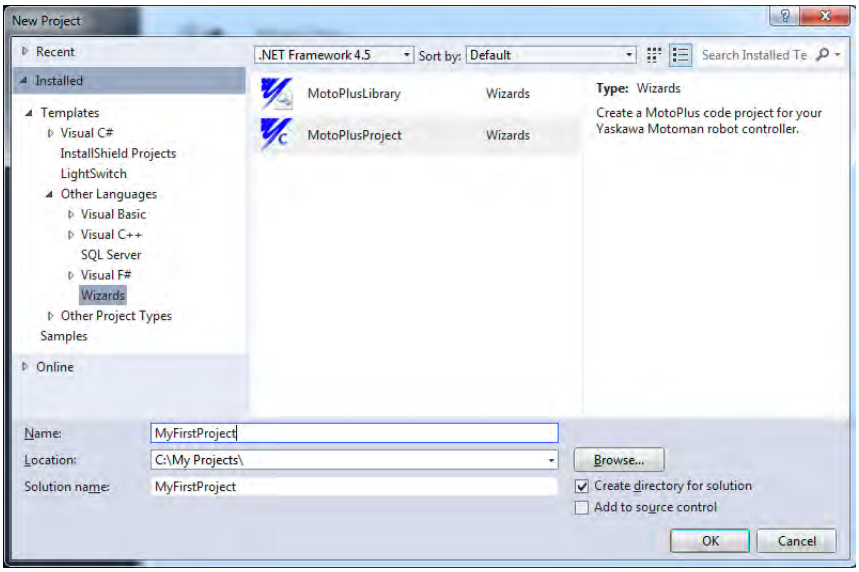
4. mpExitUsrRoot must be called at the end of mpUsrRoot.
5. The task “mpUsrRoot” is launched at a high task priority. Due to its high task priority, if a time-consuming process (0.1 milliseconds or longer) is done in this task, the processing time for the robot control becomes insufficient. This can interrupt robot startup and freeze the pendant. Therefore, you should immediately launch a new normal priority task from mpUsrRoot and then call mpExitUsrRoot.

**NOTICE**

When creating a new MotoPlus project in Visual Studio, a “main” template is automatically generated.

## 3.2 Creating a MotoPlus Project in Visual Studio

Open Visual Studio on your development PC. Select [File] > [New Project]. The MotoPlus project is found under {Templates} > {Other Languages} > {Wizards}.



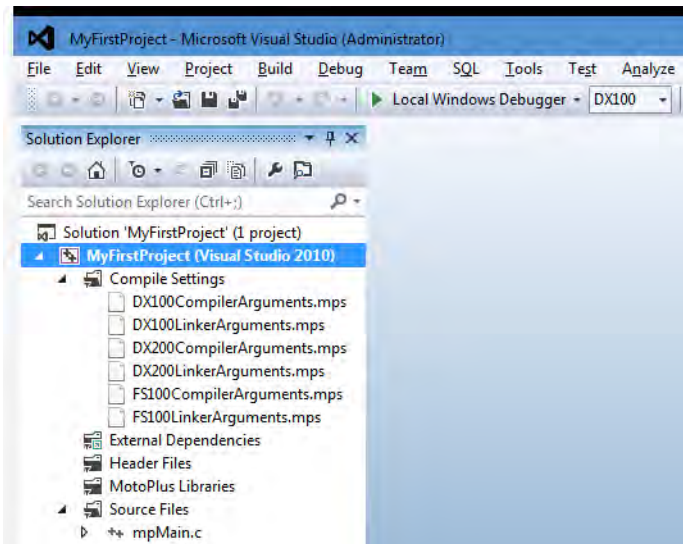

**NOTICE**

Do not include any spaces when naming your project.

Click [OK] and a new project is created in the specified location. This project is automatically configured to build using the MotoPlus compiler. A template source file (mpMain.c) is added to the project. Also included are files which specify the project-specific arguments to the compiler and linker.

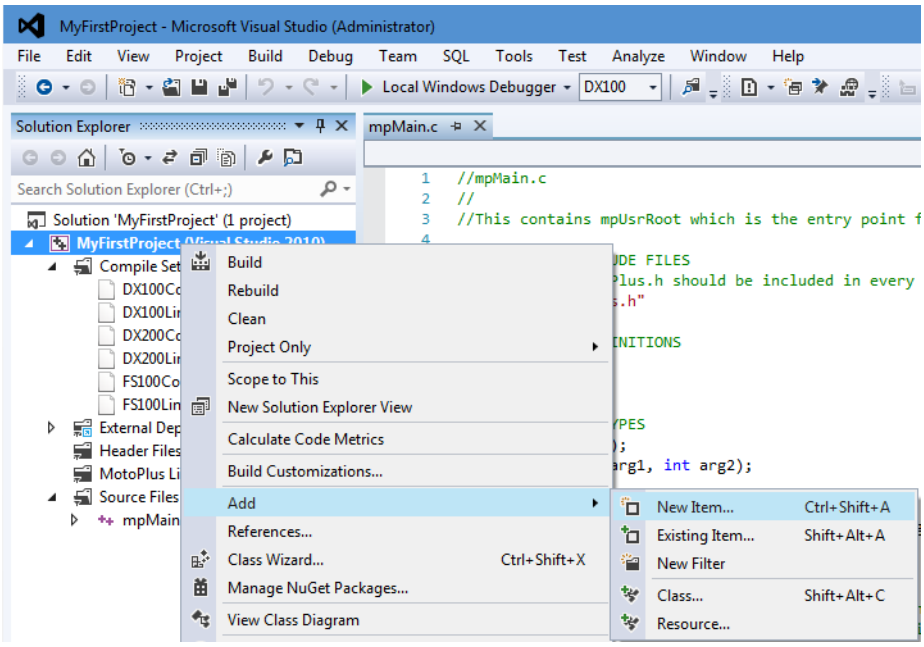
**NOTICE**

The .mps files must not be removed from your MotoPlus project.



## 3.3 Adding Files to the Project

Files can be added to your MotoPlus project through the standard Visual Studio interface. Right-click on the project tree and select {Add} > {New Item…} or {Add} > {Existing Item…}. Alternatively, you can perform the same action from the {Project} dropdown in the Main Menu.




When adding new items to the project, the file must end in a supported extension. Header files should have a “.h” extension. Source files should have a “.c” extension.

| NOTICE |
| --- |
| C++ (.cpp) files are not supported by the MotoPlus environment. |

Files for the project do not have to be in a flat directory. They can be in any folder relative to the Project Directory.

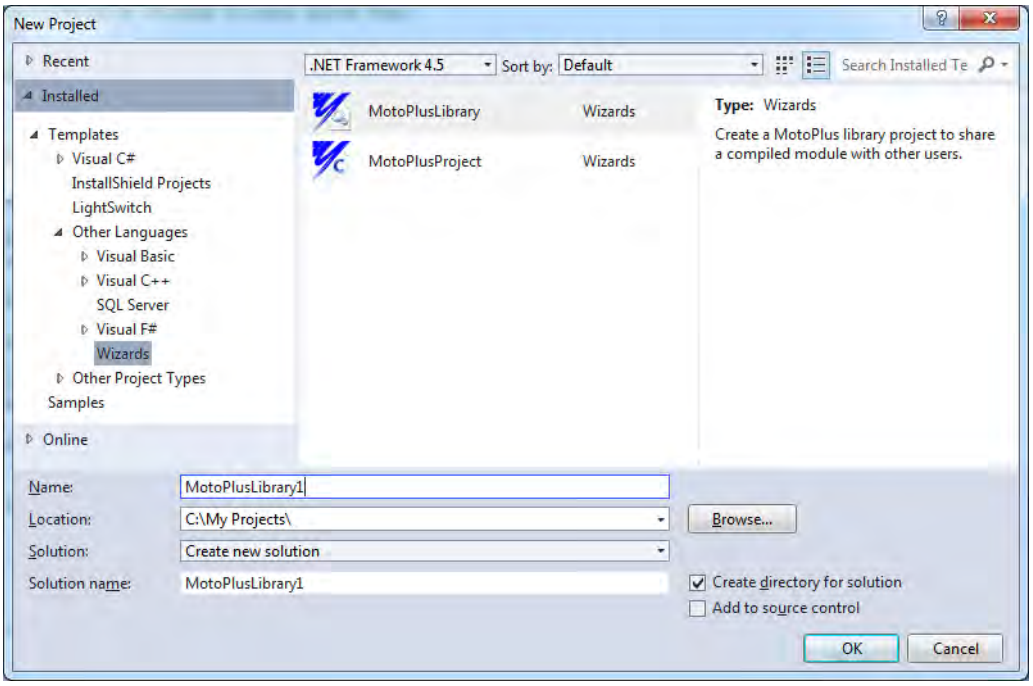
## 3.4 Creating a MotoPlus Library

As a MotoPlus developer, you can share routines by distributing a compiled MotoPlus library. This library can be used by other developers without exposing proprietary source code. Instead of building an executable module (.out), this will build a library without the main entry point for execution.

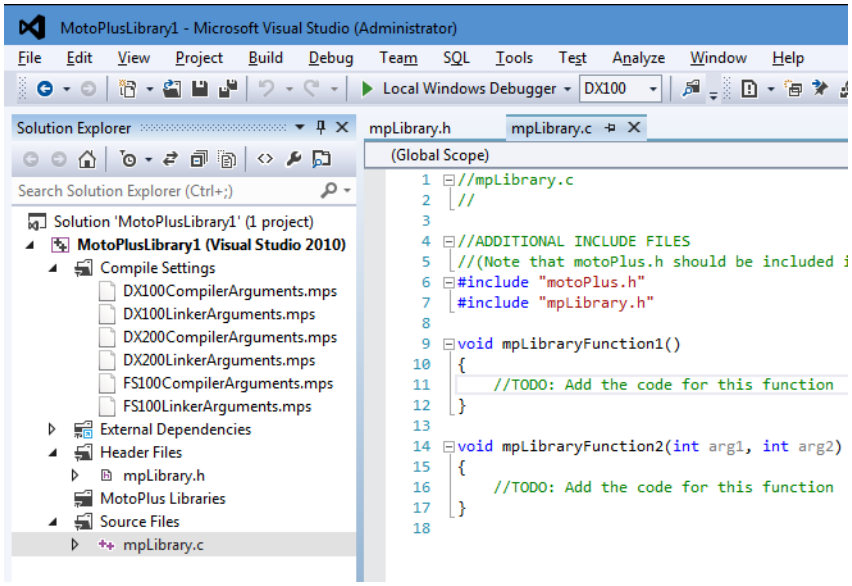
**NOTICE**

MotoPlus library files end in extension .mpLib (DX100),.fsLib (FS100), .dnLib (DX200) and .yrcLib (YRC1000 or YRC1000micro). Libraries must be compiled for each robot controller separately.

A MotoPlus library is created in the same manner as a MotoPlus project. From the New Project window, select the MotoPlusLibrary wizard.



A project is generated with a template source file (mpLibrary.c) and header file (mpLibrary.h) to get you started.




The project is automatically configured to build a library instead of executable file.

**NOTICE**

When distributing a MotoPlus library, be sure to include a header file which provides functional declarations for each public function in your library. Function declarations should begin with “extern”.

Example: `extern void` mpLibraryFunction1();

## 3.5 Using a MotoPlus Library In Your Project

MotoPlus library files can be added to your project like any other file. Right-click the project tree and select {Add} > {Existing Item…}.

The library developer should have also provided a header file with the declarations for all public functions in the library. Each declaration should begin with “extern”. Example:

```
extern void mpLibraryFunction1();
```

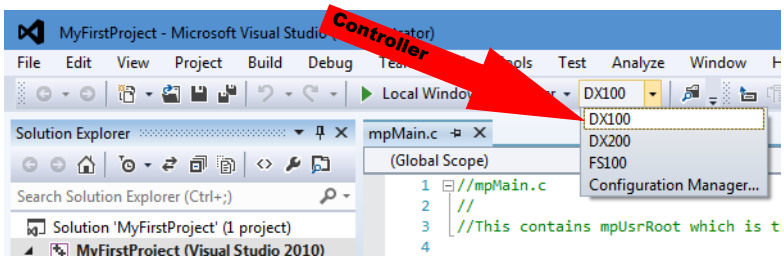
If your application is targeted toward multiple controller platforms, you must add a library for each controller type. When building the project, the compiler will ignore libraries which are targeted for a different controller platform.

| NOTICE |
| --- |
| MotoPlus library files end in extension .mpLib (DX100), .fsLib (FS100), .dnLib (DX200) and .yrcLib (YRC1000 or YRC1000micro). |

## 3.6 Building Your Application

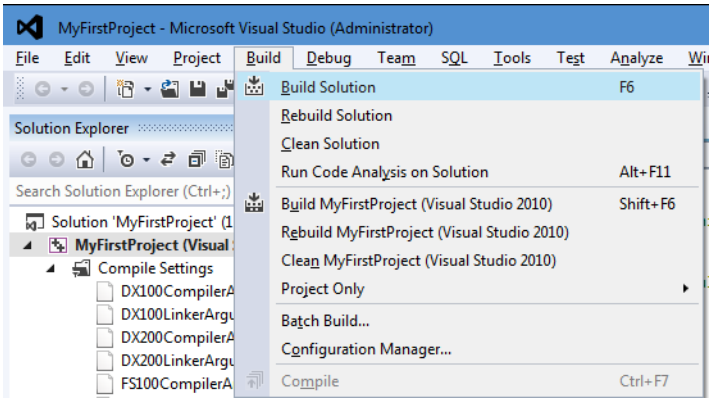
The MotoPlus application (or library) can be built directly from within Visual Studio. The project is automatically configured to pass all files in the Solution Explorer to the MotoPlus compiler.

First select your target controller platform from the Configuration Manager dropdown.

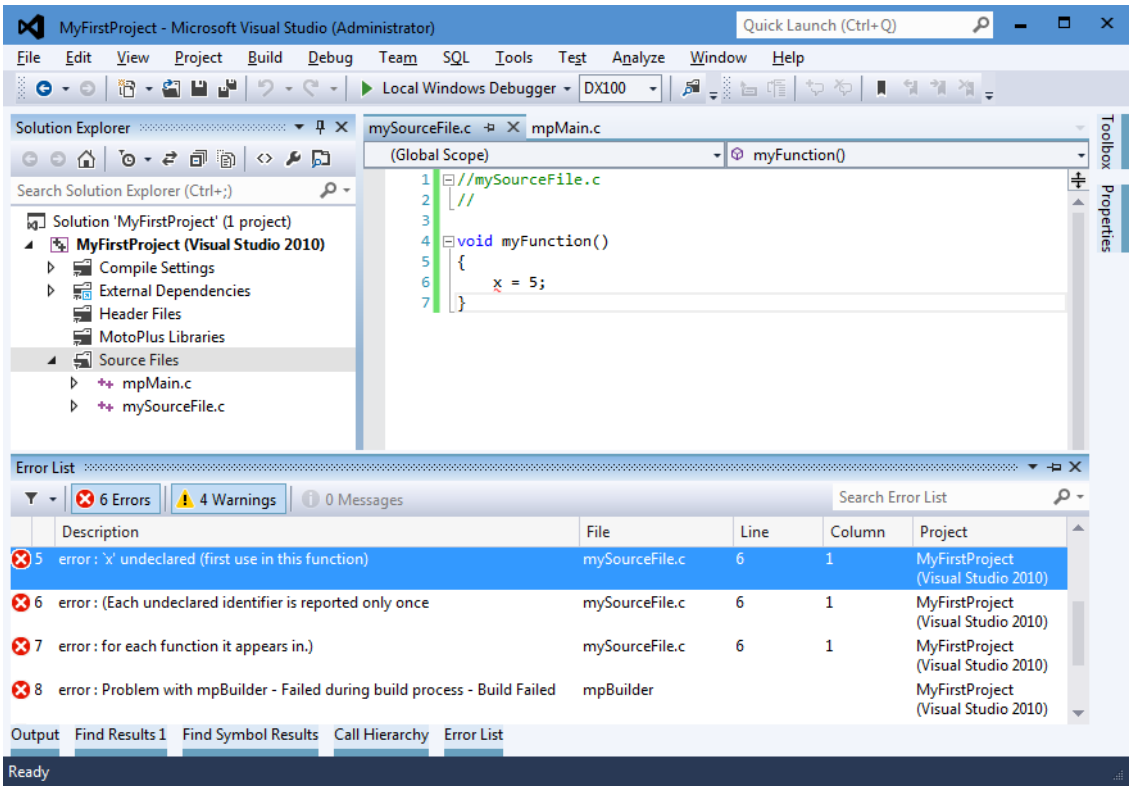



| NOTICE |
| --- |
| You must have a license key for each target controller platform. Upgrade licenses are available for each controller type. |

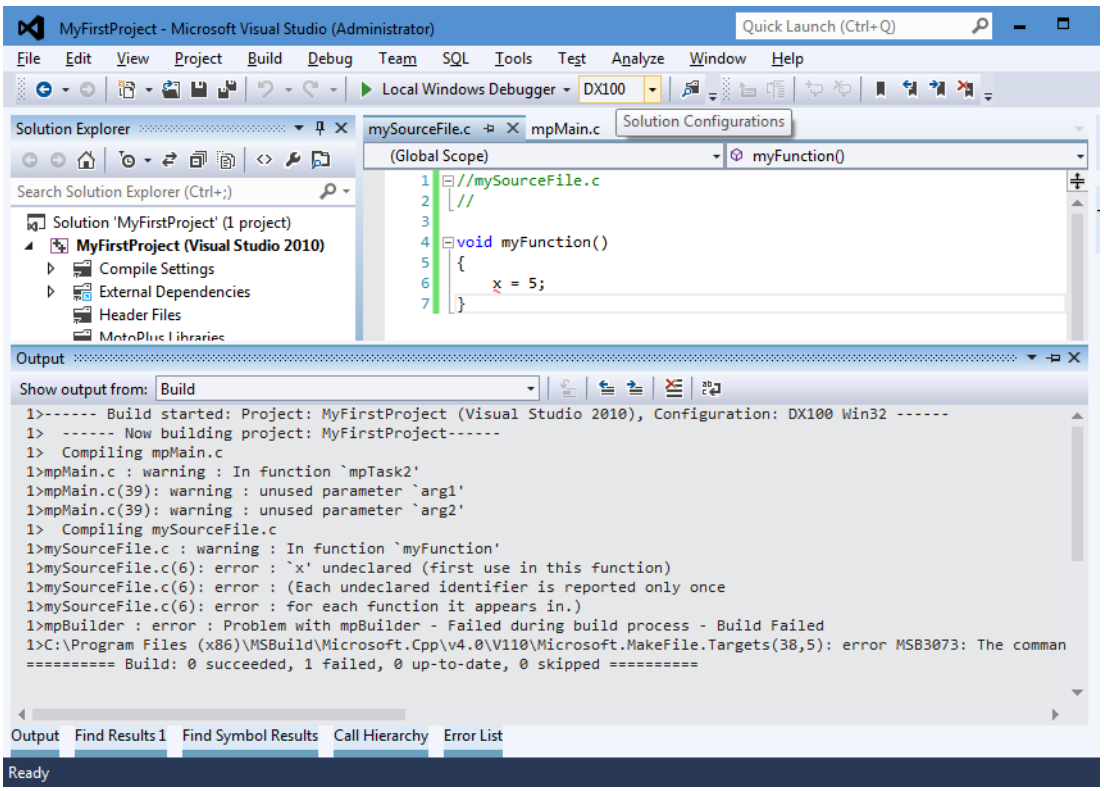
Next, select the appropriate Build option from the {Build} menu.



If there are any errors/warnings in your source files, they will be displayed in the Visual Studio Error List. You can double-click the error entry to go directly to the problematic line of code.

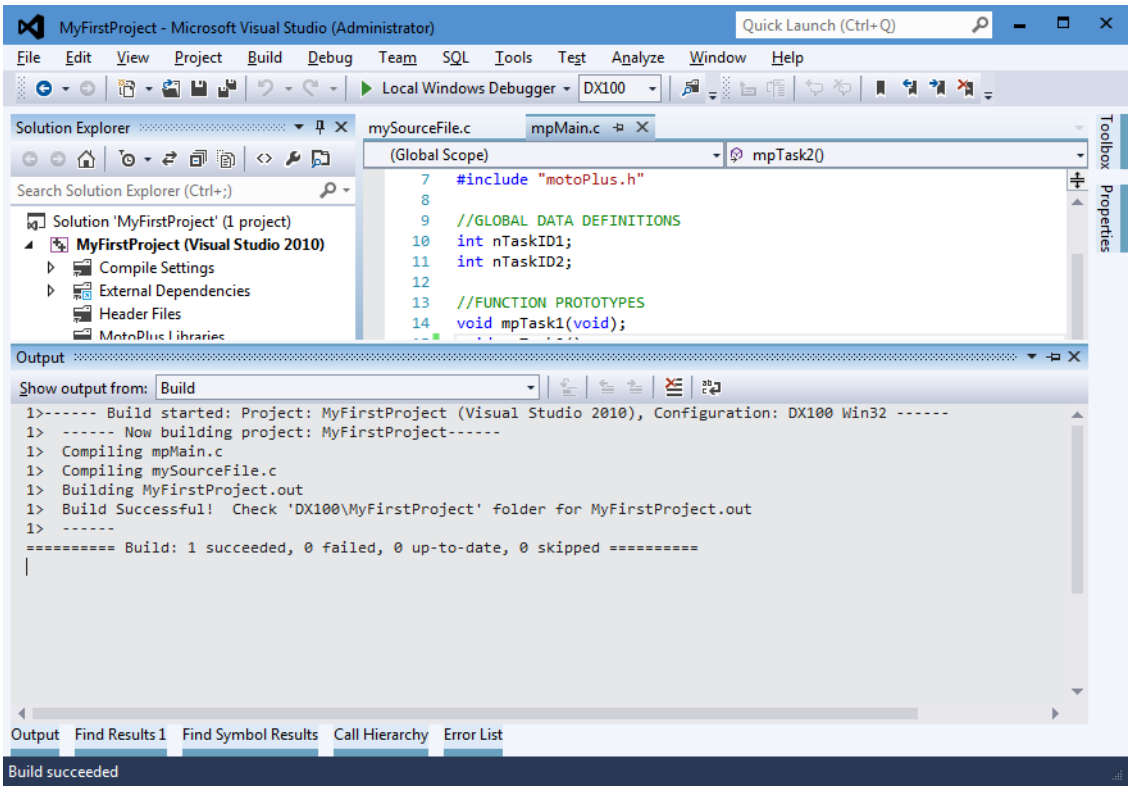



The Output pane contains the full output text of the compiler.



If there are no errors in the code, the Output pane will indicate “Build Successful!” By default, the compiled module will be located in the folder:

<Solution Directory> \ <Controller Platform> \ <Project name>




.

| NOTICE |
| --- |
| The output directory and file name can be customized by modifying the NMake Configuration Properties.<br><br>See *Chapter 7 “Advanced mpBuilder Usage”* for more details |

## 3.7 Building Your Project for Multiple Controller Platforms

The MotoPlus for Visual Studio SDK allows you to build your source code for each of the MotoPlus-enabled YASKAWA robot controllers. This allows you to use the same source code project for each controller. However, there are minor differences between each controller which may affect your application. For example, the Network Inputs I/O addresses start at #25010 for DX100 and FS100. However, the starting address is #27010 on the DX200, YRC1000, and YRC1000micro.

For this reason, the build environment has been configured to automatically provide a pre-processor definition for each controller type. Using preprocessor directives, you can put a decision statement around your code at compile-time, instead of run-time.

Example:

```
//evaluated at compile-time
#if (DX100 || FS100)
   #define STARTING_NETIN_ADDRESS     25010
#elif (DX200 || YRC1000 || YRC1000micro)
   #define STARTING_NETIN_ADDRESS     27010
#endif

void myFunction()
{
   //set at run-time
   int startingIoAddress = STARTING_NETIN_ADDRESS;
}
```

**NOTICE**

You must have a license key for each target controller platform. Upgrade licenses are available for each controller type.

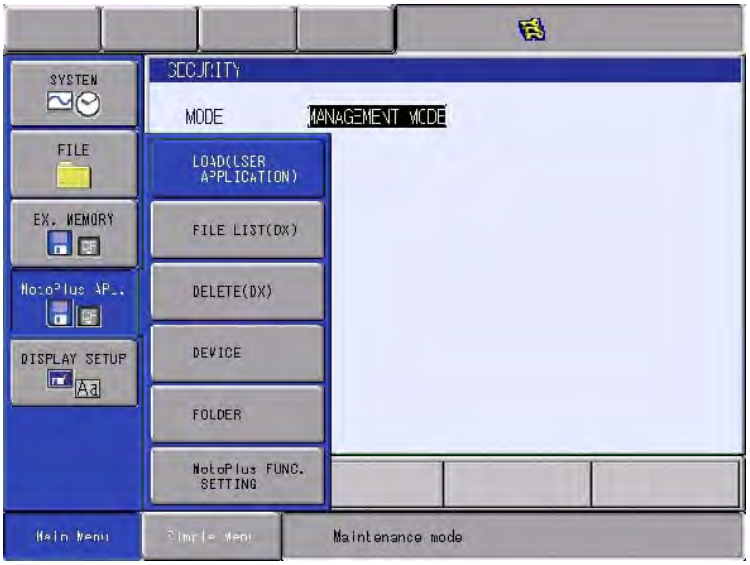
# 4 Installation and Startup of MotoPlus Application

## 4.1 Installation of Application Program in Maintenance Mode

1. Load Function
   Load the application program and install by following the instructions below.

   (1) Boot the robot controller into MAINTENANCE mode by holding [Main Menu] on the programming pendant.

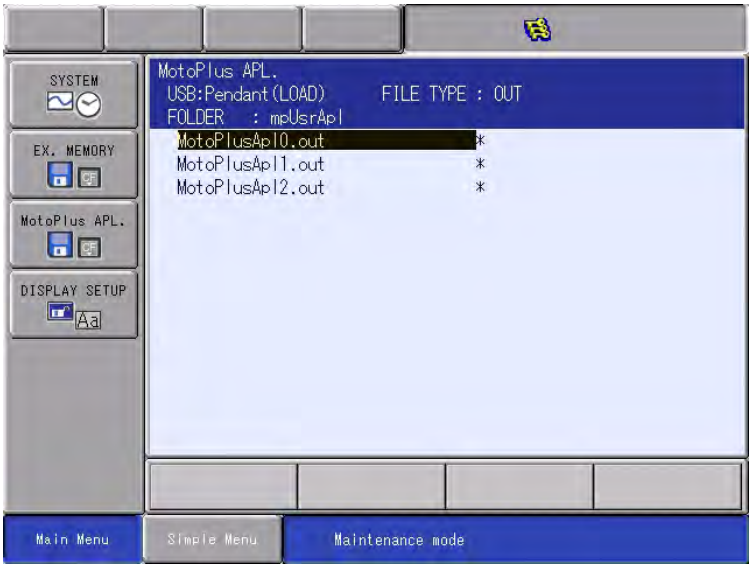
   (2) Set the security mode to MANAGEMENT mode.
   Select the Main Menu {MotoPlus APL.}. Then {LOAD (USER APPLICATION)}, {FILE LIST}, {DELETE}, {DEVICE}, {FOLDER}, and {MotoPlus FUNC. SETTING} appear as submenus.



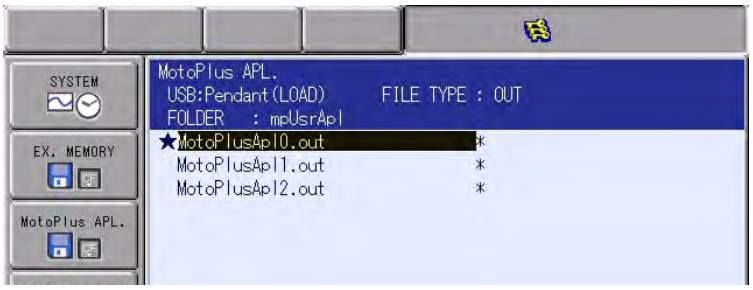

   (3) Specify the file location.
   Select an appropriate file location with the submenus {DEVICE (to select CF/USB)} and {FOLDER (to select the folder which has the application program)}.

(4) Select and load a file.

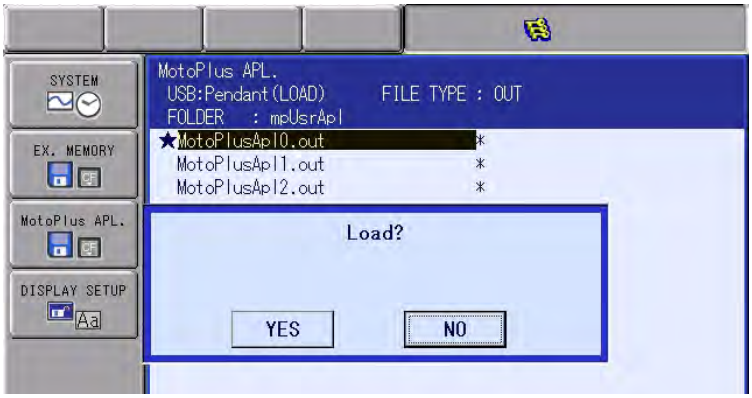
• Select the submenu {LOAD (USER APPLICATION)}. Then, "*.out" files in the specified folder appear as shown below.



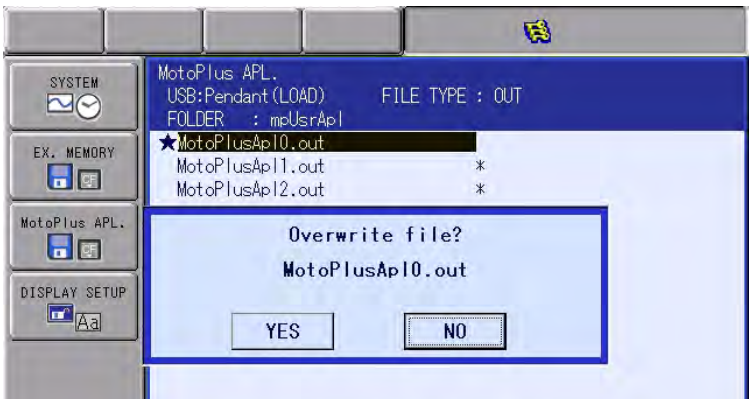

• Move the cursor and press [SELECT]. Then, the selection indicator "★" appears on the left of the application file name. Press [SELECT] again to clear "★".



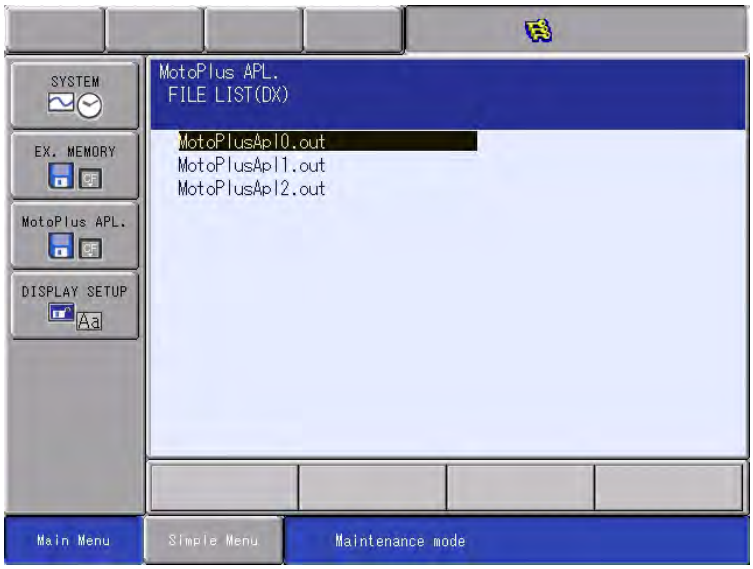

• Press [ENTER], then the following confirmation dialog box appears.



• When {YES} is selected and the selected application file contains the file which already exists in the controller, the following confirmation dialog box appears. Select “YES” to load.

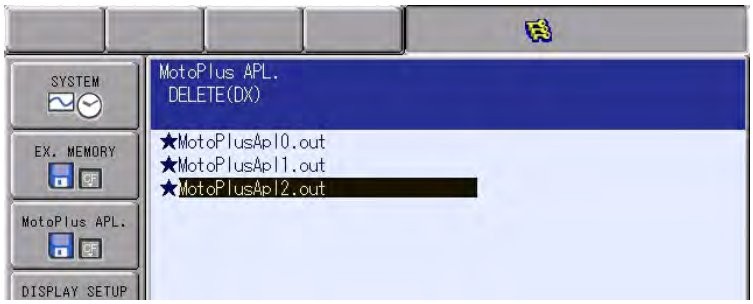



2. List
Select the submenu {FILE LIST}. Then, the list of the application files which already exist in the controller appears.

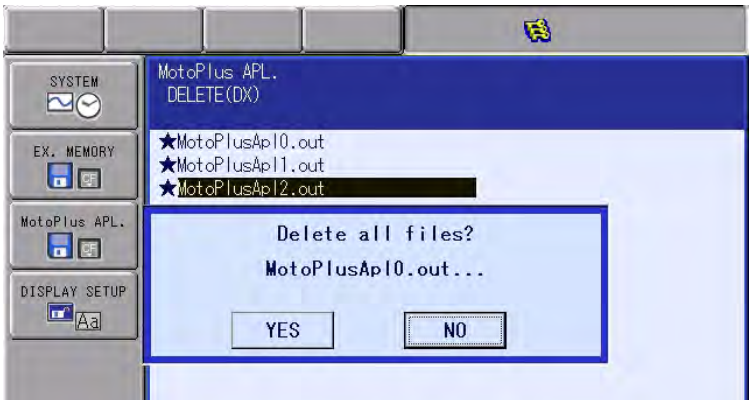



3. Deletion of application program
By the submenu {DELETE}, the application file which already exists in the controller can be deleted.

(1) Move the cursor and press [SELECT]. Then, the selection indicator “★” appears on the left of the application file name. Press [SELECT] again to clear “★”.



(2) Press [ENTER], then the following confirmation dialog box appears.



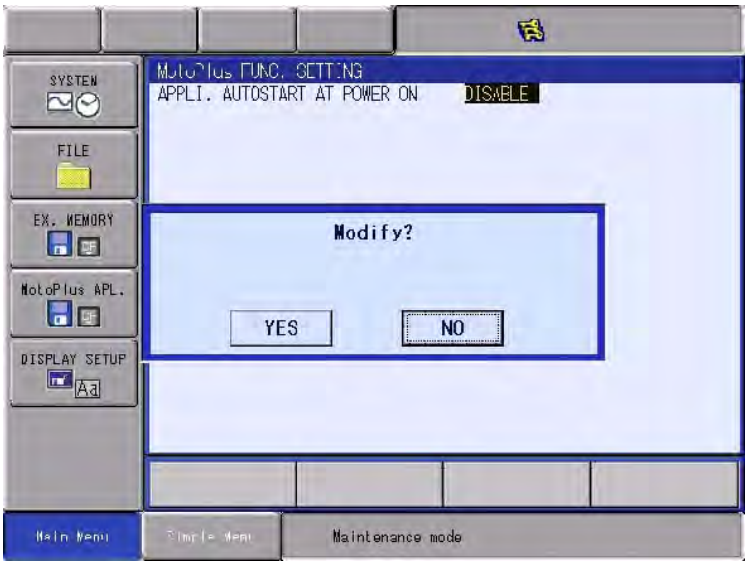

4. MotoPlus Function Setting
When selecting “MotoPlus FUNC. SETTING”, the following windows appear to enable or disable the autostart of the MotoPlus application program. For the detailed usage methods, refer to *chapter 4.2 “Start-up of Application Program” on page 4-6* and *chapter 4.3 “To Temporarily Prevent MotoPlus Application from Starting” on page 4-6*.

(1) Move the cursor and press [SELECT]. The setting is changed from “ENABLE” to “DISABLE”.

**NOTICE**

This screen may have different options based on which robot controller is being used. Please refer to the Programmer's Manual for more details.

(2) Press [ENTER], and then the confirmation dialog box appears.




(3) Select {YES}.

–When {YES} is selected on the confirmation dialog box, the parameter is set automatically, and then it returns to the MotoPlus setting window.

## 4.2 Start-up of Application Program

When the controller is turned ON, the application program is loaded automatically and started.

**<Limitations>**

- Only one “*.out” file can be loaded. Thus, install only one application. If two or more applications are installed, the alarm 1020: “MOTOPLUS APPLICATION LOAD ERROR [1]” occurs on loading. The error status can be checked by reading $B051 from the job.
- The loadable memory size is up to 2 Mbyte as the total of the code area and the static memory area. If it is more than 2 Mbyte, the start-up fails and the alarm 1020: “MOTOPLUS APPLICATION LOAD ERROR [4]” occurs on loading. The error status can be checked by reading $B051 from the job.

  $B051 0: The application is loaded successfully on start-up.
  1: The number of files exceeds the limit.
  2: The memory is insufficient. (Available memory area is less than 2 Mbyte.)
  3: APPLICATION folder cannot be found in the CF.
  4: The size of the MotoPlus application exceeds the limit.

## 4.3 To Temporarily Prevent MotoPlus Application from Starting

Due to a problem of the application program, the system may hang up during robot operation, and then the controller may not start normally. For recovery, it is necessary to start the controller normally while preventing the application program from operating. In this case, it is necessary to start the controller in the MAINTENANCE mode, change the settings so that the application program does not operate, start the controller normally, and operate the robot. Follow one of the following two procedures:

1. Start the controller in the MAINTENANCE mode, set the security mode to the MANAGEMENT MODE. Select “MotoPlus APL.”, then “DELETE” to delete the application.
2. Start the controller in the MAINTENANCE mode, set the security mode to the MANAGEMENT MODE. Select “MotoPlus APL.”, then “MotoPlus FUNC. SETTING” to show the setting window. Then set “APPLI. AUTOSTART AT POWER ON” to “DISABLE”.

*If the above problem occurs during robot operation, choose procedure* 2. The above problem during robot operation may be caused by a specific condition. If the MotoPlus application is deleted with procedure 1, after the specific condition is cleared, the MotoPlus application must be loaded again in the CF for recovery. In this case, if the application to be loaded does not exist there, the application cannot operate and the system cannot recover.

# 5 Debugging

## 5.1 Debugging Environment

Due to the nature of the real-time robotic system, MotoPlus applications cannot be debugged using a traditional source-level debugger. The most effective way to debug a MotoPlus application is to monitor application output. This is done by setting variables and I/O to view on the pendant, or by printing to standard output (stdout) within your application.

Print-statements sent to stdout can be viewed by using one of two methods. On DX and YRC1000 robot controllers, your SDK comes with a VGA adapter cable to connect a standard computer monitor to the robot controller. Alternatively, you can connect to the robot via telnet to view output within a console window on your PC.

| NOTICE |
| --- |
| Print-statements are extremely time consuming during execution. It is highly recommended to put a debug-switch around all print statements so that they can be disabled for final release. |

## 5.2 Commands for Printing to Standard-output (stdout)

The commands “puts” and “printf” are available to print to stdout. The command “puts” will print a string and automatically append a new line. The command “printf” is used to print a formatted string with variable arguments. The line-feed is not automatically appended to printf strings.

## 5.3 Viewing Output on Computer Monitor

**(DX100, DX200 and YRC1000)**

Please see *chapter 2.3.1 “Install VGA Adapter Cable for Debugging” at page 2-3* for instructions on installing the VGA adapter cable included with the MotoPlus SDK.

To view output using the monitor, the monitor must be connected prior to turning on the robot controller. If the monitor is blank, please cycle power on the robot controller.

## 5.4 Viewing Output on PC Console Window

### 5.4.1 Setup of Robot Controller

Please see *chapter 2.4 "Enable Telnet for Debugging" at page 2-7* for instructions on enabling the Windows Telnet Client in Windows 7.

Set robot parameter S2C1119 = 2 to enable telnet output. The robot must be rebooted after setting this parameter.

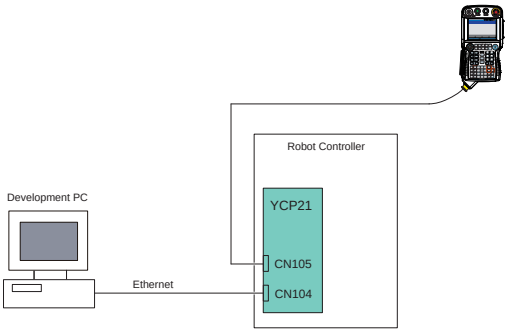
**NOTICE**

When debugging is completed, be sure to set S2C1119 = 0. This will prevent the telnet task from interfering with robot operation.

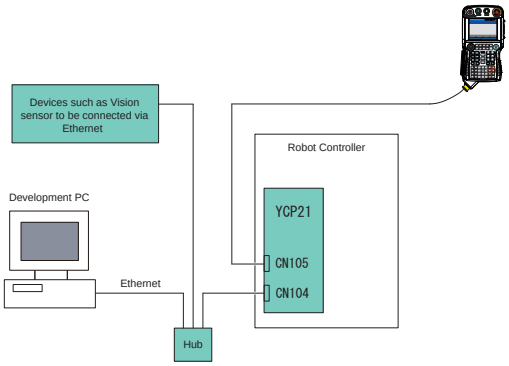
Connect your development PC to the robot controller LAN port via Ethernet connection.

■ **DX and YRC1000 Controller**

**Without a Hub**

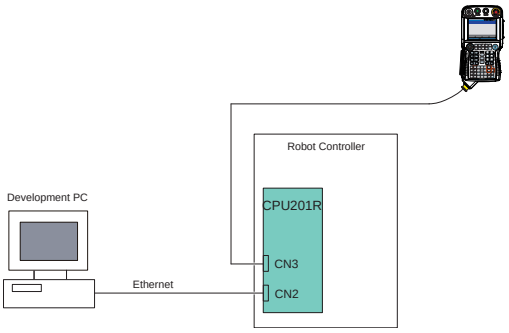



**With a Hub**

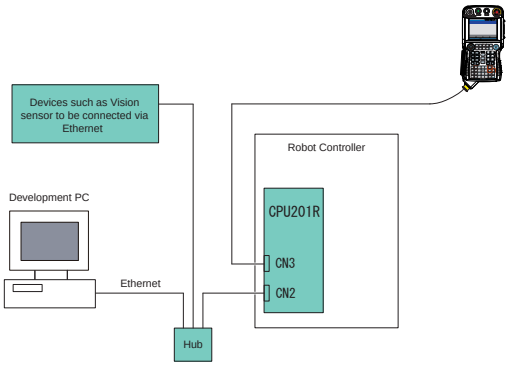



■ **FS Controller**

**Without a Hub**




**With a Hub**



### 5.4.2 Establishing Telnet Connection

**NOTICE**

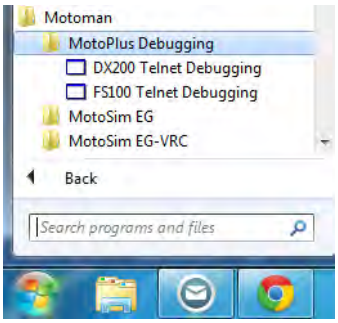
- Your computer IP address must be on the same subnet as the robot controller.
  - Example:
    Robot IP - 192.168.255.1
    Computer IP - 192.168.255.xxx
- Alarms such as “0500 SEGMENT PROC NOT READY” may occur when using the telnet connection. This is because the telnet task runs at a high priority in the system and can interfere with robot operation. If the alarm occurs, reboot the robot controller.

Set S2C1119 = 0 when you are done debugging to prevent these alarms during normal operation.

#### 5.4.2.1 Automatic Login to Telnet Shell

(Not available on DX100)

On the controller, an automatic login utility is provided with the MotoPlus for Visual Studio installation. This is accessed from the START menu under Motoman\MotoPlus Debugging.




Prior to opening the automatic login utility, you must input the IP address of the robot controller. An .ini configuration file is located in your MotoPlus for Visual Studio installation directory. (Default: C:\Program Files (x86)\Motoman\MotoPlus for Visual Studio\Telnet Debugging)

Open the .ini file for your target controller. Modify the value of “ipaddress” to match your robot controller. After saving this file, you can now open the automatic login utility.

After opening the automatic login utility, do not click any other windows until the login process is complete. The process is completed when you see the telnet shell prompt “->”.

#### 5.4.2.2 Manual Login to Telnet Shell

(all controllers)

As an alternative to the automatic login utility, you can manually login to the telnet shell.

Open a Windows console window. The easiest method is through the “Run…” command in the START menu.

START > Run… > Type “cmd”

Type “telnet” to open the Windows Telnet Client utility. Then type “open <ipaddress>” to open a connection to the robot. It will prompt you for the VxWorks login information. You have 30 seconds to type in the login username and password before it times out.

| Controller | Username | Password |
|---|---|---|
| DX100 | Net_Maint_Mng | 99999999 |
| DX200 | MOTOMANrobot | MOTOMANrobot |
| FS100 | MOTOMANrobot | MOTOMANrobot |
| YRC1000 | MOTOMANrobot | MOTOMANrobot |
| YRC1000micro | MOTOMANrobot | MOTOMANrobot |

If the login is successful, the console window will display the shell prompt “->”.

## 5.5 Development Support Tools for YRC1000 and YRC1000micro

These tools function with the YRC1000 and YRC1000micro controller.

### 5.5.1 Introduction

This chapter explains the functions of the tools to support development of the MotoPlus application (hereafter referred to as “the application”).

### 5.5.2 Overview of Functions

The MotoPlus application development support tool contains the following functions.

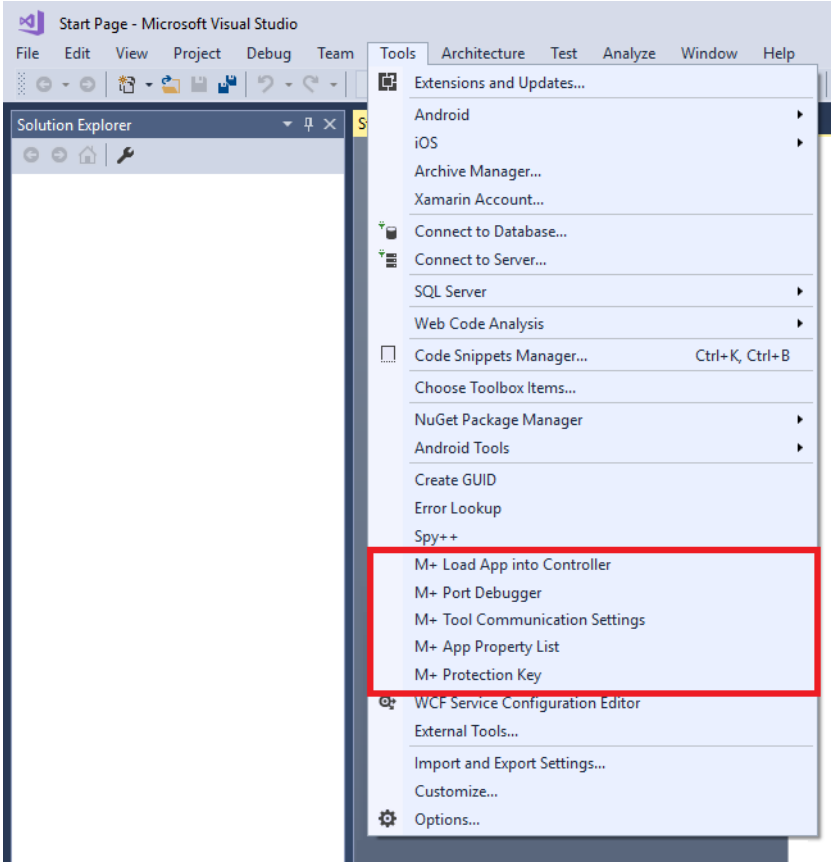
| No. | Function | Details |
|---|---|---|
| 1 | Online Download | A function which downloads the application from Visual Studio to the robot controller (hereafter referred to as “the controller”). |
| 2 | Application Property List | A function which displays the detailed information, such as the operation status of the application. |
| 3 | Port Debug | A function which measures the processing time of the application. |
| 4 | Copy Protection | A function which prevents applications that are copied illegally from being used. |

### 5.5.3 Using the Support Tools

(Visual Studio)

Each function is available from the “Tools” menu in Visual Studio. These tools are also accessible from the Windows Start menu.

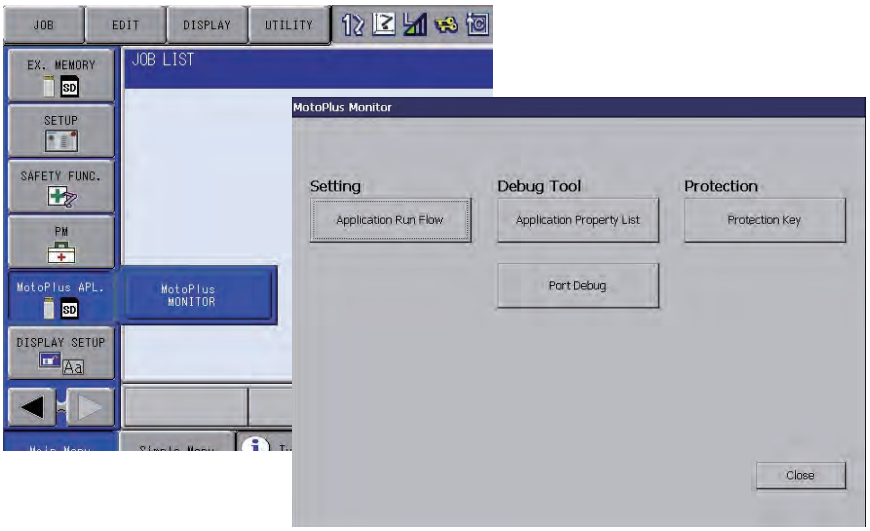
*Fig. 5-1: External Tools in Visual Studio*




(MotoPlus Monitor)

On the programming pendant from Main Menu select {MotoPlus APL.} and then select {MotoPlus MONITOR} to use the functions.

*Fig. 5-2: MotoPlus Monitor*



### 5.5.4 Basic Settings

#### 5.5.4.1 Connecting a PC to Robot Controller

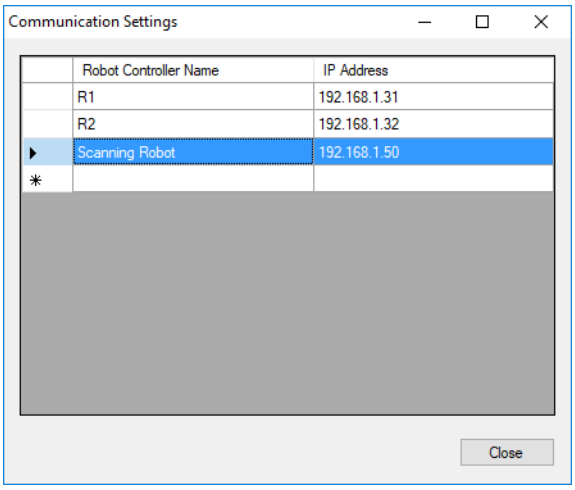
Connect the development PC to the Robot Controller with a standard Ethernet cable. Depending on which network interface(s) is enabled on the controller, the PC should be connected to CN106 (LAN2) or CN107 (LAN3). If other devices are already connected to the network interface, then an Ethernet switch may be used.

#### 5.5.4.2 Communication Settings

When using any of the development support tools, you must specify the robot to connect to via IP address. The addresses for each of the robot controllers must first be registered using the Communication Settings tool.

From within the Communication Settings tool, simply type a name and IP address for each robot that you will be connecting to. This list of robots will be selectable within each of the development support tools.

*Fig. 5-3: Communication Settings*



### 5.5.5 Online Download

This function downloads the application from the development PC to the robot controller.

Loading the application is usually performed in the maintenance mode, however, this function allows the application to load while the controller is online. Since this function bypasses the time and effort required for restarting in maintenance mode, the operating time can be reduced when debugging the application. A USB storage drive is required when using this function.

*Fig. 5-4: Online Download*

### 5.5.6 System Settings

The following parameter is required to enable the Online Download function.

**NOTICE**

Use this function only when debugging the application.

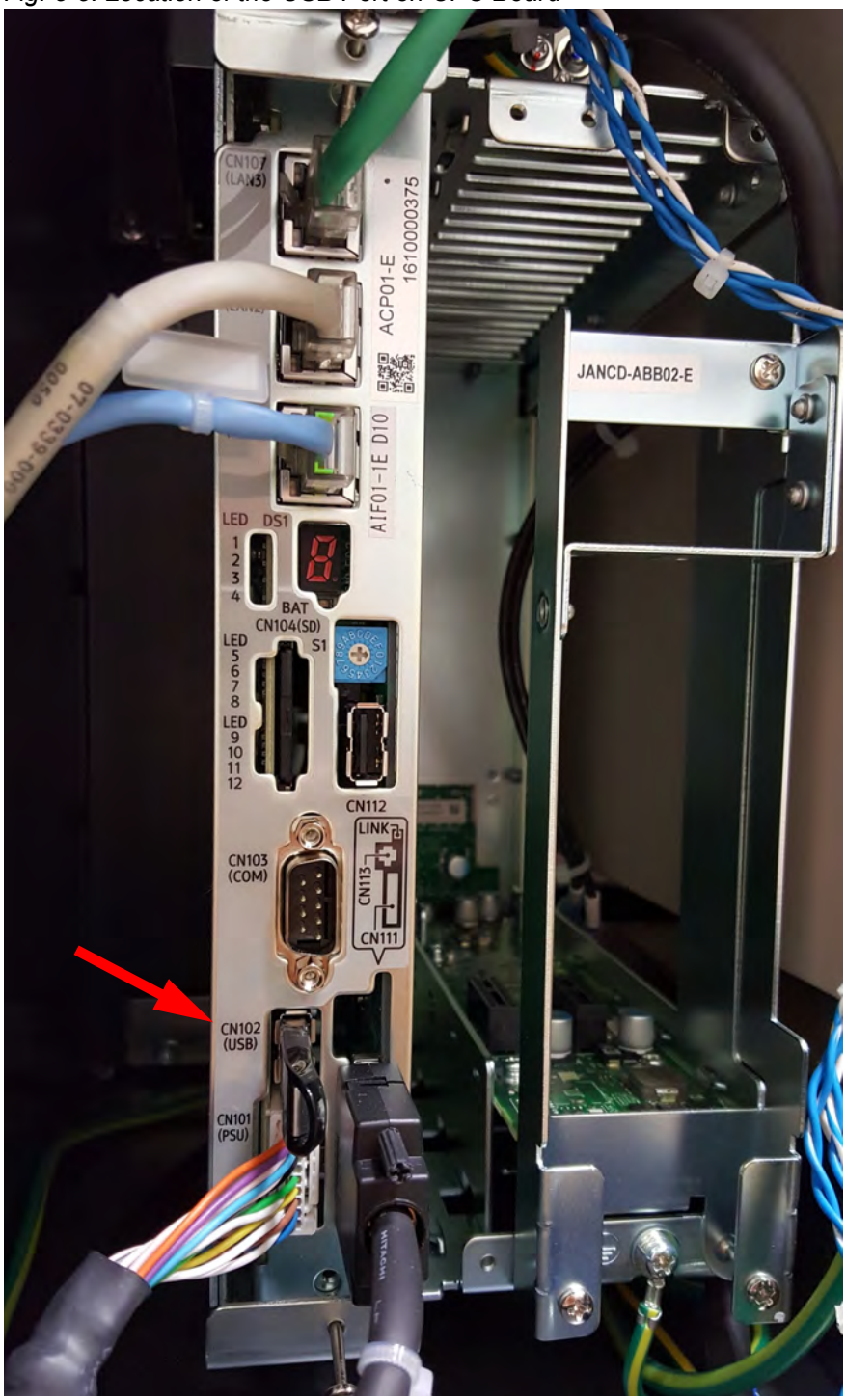
When the application is released, return the parameter to the original value, and then load the application from within maintenance mode as usual.

| Parameter | Setting value | Details |
|---|---|---|
| S2C1113 | 1 | Enable Online Download function<br>0: Invalid, 1: Valid |

### 5.5.7 Inserting the USB Memory in the Controller

This function allows the application file to be saved to the controller. Attach the flash drive to the USB port on the CPU board.

*Fig. 5-5: Location of the USB Port on CPU Board*



### 5.5.7.1 Operating Procedure

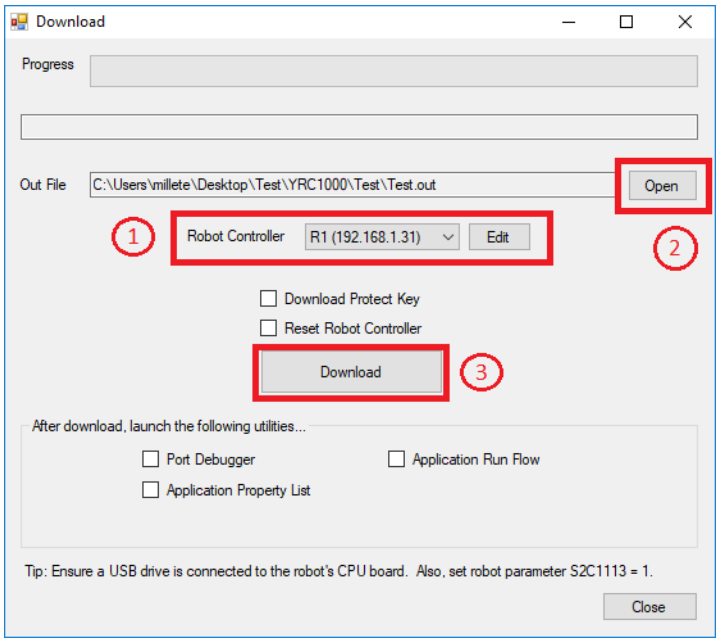
From the “Tools” menu in Visual Studio, select “M+ Load App into Controller”.

**NOTICE**

This tool automatically launches when selecting “Start Debugging” [F5].

1. Select the controller to receive the download.
2. Select the application file (.out file) to download.
   This field automatically populates with the path of the application file which was compiled last.
3. Click [Download].
4. Turn OFF the controller and then turn it back ON.
   If the “Reset Robot Controller” option is selected, the controller automatically restarts when the download completes.

*Fig. 5-6: Online Download Window*



### 5.5.8 Application Property List

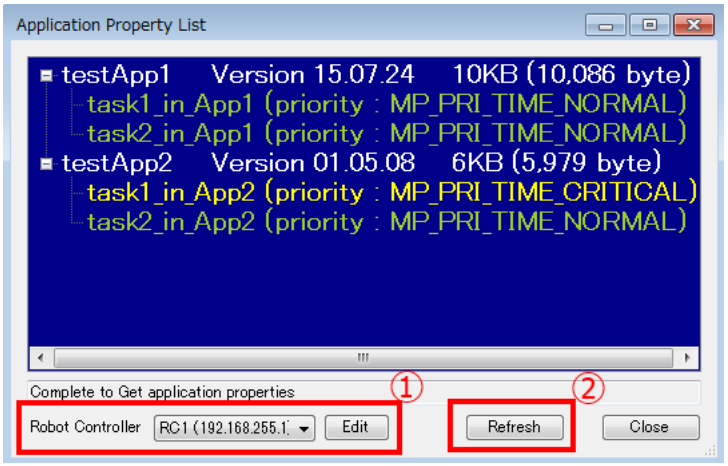
Acquires and displays the application properties. The following properties can be displayed.

- Application name
- Application version
- Application file size
- Task name
- Task executing priority

#### 5.5.8.1 Tool Operating Procedure:

- **Using Development PC:** select “M+ App Property List” from the “Tools” menu in Visual Studio.
- **Using Programming Pendant:** select {MotoPlus APL} > {MotoPlus Monitor}. Then choose {Application Property List}.

*Fig. 5-7: Application Property List Window*




#### 5.5.8.2 How to Set Application Information

The following code example demonstrates how to set the Application Information data.

*Fig. 5-8: Setting the Application Information*

```
void mpUsrRoot(int arg1, int arg2, int arg3, int arg4, int arg5, int arg6, int
{
    int taskId;
    MP_APPINFO_SEND_DATA appInfo;
    MP_STD_RSP_DATA rData;

    sprintf(appInfo.AppName, "testApp1");
    sprintf(appInfo.Version, "15.07.24");
    sprintf(appInfo.Comment, "Test");

    mpApplicationInfoNotify(&appInfo, &rData);

    taskId = mpCreateTask(MP_PRI_TIME_NORMAL, MP_STACK_SIZE, (FUNCPTR)mpTask3,
            arg1, arg2, arg3, arg4, arg5, arg6, arg7, arg8, arg9, arg10);
    mpTaskPropNameSet(taskId, "task1_in_inApp1");

    taskId = mpCreateTask(MP_PRI_TIME_NORMAL, MP_STACK_SIZE, (FUNCPTR)mpTask4,
            arg1, arg2, arg3, arg4, arg5, arg6, arg7, arg8, arg9, arg10);
    mpTaskPropNameSet(taskId, "task2_in_inApp1");

    mpExitUsrRoot;
}
```

### 5.5.8.3 Setting Procedures

1. Set the application name and the version by using mpApplicationInfoNotify(). Use the MotoPlus reference manual for details on mpApplicationInfoNotify().
2. Set the task name by using mpTaskPropNameSet().

# mpTaskPropNameSet

Set the task name to be displayed in the Application Property List.

■ **Syntax**

```
int mpTaskPropNameSet(
int taskId,
char* name
);
```

■ **Argument**

[taskId]

Specify the task ID that is returned by using mpCreateTask().

[name]

Specify the task name. For the task name, use a maximum of 32 characters.

■ **Return value**

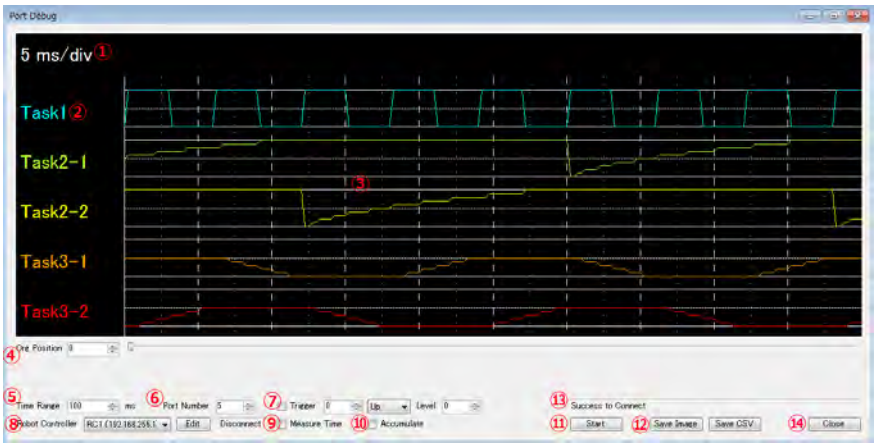
0: Normal end

-1: Error

| NOTICE |
| --- |
| When the information is not set, the default setting value is displayed. |

### 5.5.9 Port Debug

The Port Debug tool measures code execution, similar to an oscilloscope function. The data for this function is set using special MotoPlus API functions for debugging (mpDebugPortXXXX).




| No. | Name | Details |
|---|---|---|
| ① | Time Division | Shows the unit [ms/div] of the time axis (the horizontal axis). |
| ② | Port 0 - Port 4 | Shows the measuring port. Any name can be set by using the debug port API (for details, refer to *section 5.5.9.3 “Port Debug”*). |
| ③ | Graph Area | Shows the measuring data. Displays the time which is set in the Time Range. |
| ④ | Org Position | Optional for measuring.<br>Sets the home position of the measuring data.<br>(For details refer to *section 5.5.9.2 “Port Debug”*.) |
| ⑤ | Time Range | Sets the displaying time [ms] for measuring data. |
| ⑥ | Port Number | Sets the number of the measuring port. |
| ⑦ | Trigger | Optional for measuring.<br>Measures by applying the trigger.<br>(For details refer to *chapter 5.5.9.2.*) |
| ⑧ | Robot Controller | The controller to use this function |
| ⑨ | Measure Time | Optional for measuring.<br>Measures the time.<br>(For details refer to *chapter 5.5.9.2.*) |
| ⑩ | Accumulate | Optional for measuring.<br>Accumulates the measuring data.<br>(For details refer to *chapter 5.5.9.2.*) |
| ⑪ | Start (Stop) | Starts/stops the measuring. |
| ⑫ | Save Image / Save CSV | Saves the measuring data. |
| ⑬ | Status Bar | Shows the operating status. |
| ⑭ | Close | Port Debug exits. |

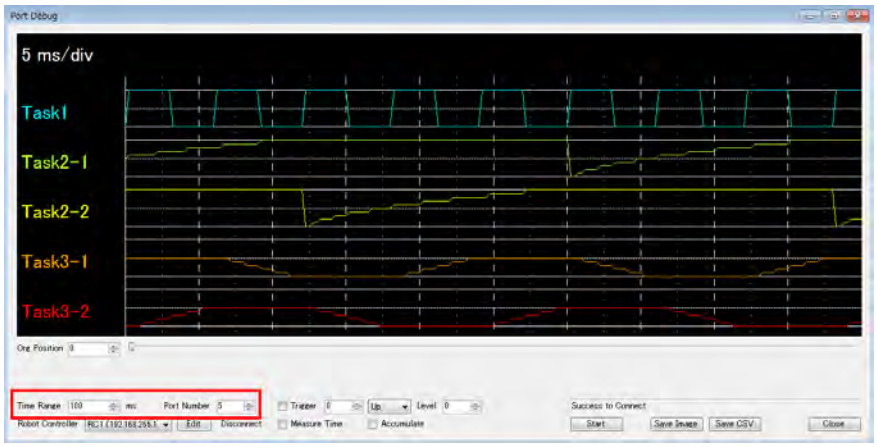
### 5.5.9.1 Measuring Procedures

From the “Tools” menu in Visual Studio, select “M+ Port Debug”.

This function can also be used on programming pendant. Select {MotoPlus APL} > {MotoPlus Monitor}. Then choose {Port Debug}.

1. Set the measuring conditions.
   Time Range Sets the displaying time [ms] for measuring data.
   Port Number Specify the number of the measuring port.
2. Press {Start} to start measuring in Port Debug. After starting, switch {Start} to {Stop}. Press {Stop} to finish measuring.

*Fig. 5-9: Port Debug Window*




### 5.5.9.2 Optional for Measuring

Trigger If the Trigger is enabled, data is only displayed when the trigger conditions are satisfied.
Set the port number and trigger conditions to apply the trigger. The trigger conditions are shown below.

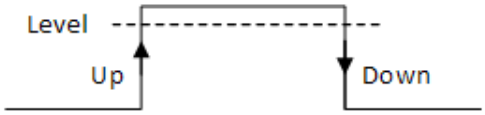
Up Detects the rising edge.

Down Detects the falling edge.

Up&Down Detects both edges, rising and falling.

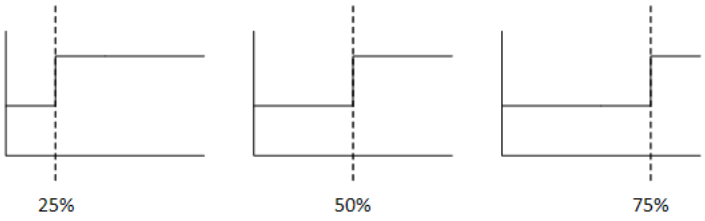
Level Detects exceeding of the threshold level.

*Fig. 5-10: Trigger*



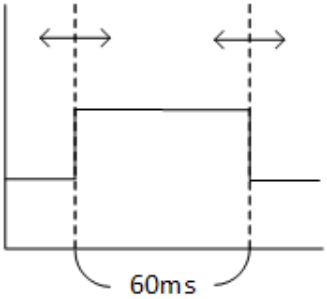
Org Position — Adjusts the display of the home position of the measuring data.
Depending on the adjustment of the home position, the displays differ as shown below.
(When the trigger is “UP”)

*Fig. 5-11: Home Position*



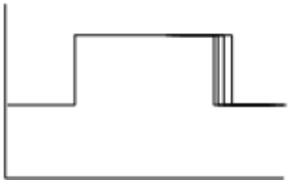

Measure Time — When this option is enabled, two cursors display for measuring a time lapse in the captured data.
When adjusting the position of the cursors, any time can be measured.

*Fig. 5-12: Measure Time*




Accumulate — Accumulates the measuring data. When this option is enabled, all measuring data is drawn by overlapping. The example below shows when the option is enabled. (Trigger is Up.) The deviation of the timing when the port is turned to 0 is caused by the deviation of the position of the rising edge.

*Fig. 5-13: Accumulate*



Save Image / CSV The measuring results are saved as an image file or a CSV file.

For Visual Studio, specify the destination to save and the file name, and then press the save button.

For MotoPlus Monitor, specify the external memory device to save to, and then press the save button.

The results are saved in the root folder of the selected external memory device.

The name of the image file is “PortDebugImageYYYYMMDD_HHMMSS.bmp”.

The name of the CSV file is “PortDebugImageYYYYMMDD_HHMMSS.csv”.

YYYYMMDD_HHMMSS indicates the time stamp when saving (Year/Month/Date/Hour/Minute/Second).

#### 5.5.9.3 Registering Data for the Port Debug Function

Before using the debug port function, it is necessary to execute mpDebugPortNameSet() once in the application.

The number of ports which are available is 5. It is possible to set values from 0 to 100 for each port.

*Fig. 5-14: How to Implement Port Debug*

```
void mpTask2(void)
{
    mpDebugPortInit(1, "Task2-1");
    mpDebugPortInit(2, "Task2-2");

    FOREVER
    {
        int i;

        for (i = 0; i <= 10; i++)
        {
            mpDebugPortLevelSet(1, i * 10);
            mpTaskDelay((i + 2) / mpGetRtc() / 2);
        }
        for (i = 0; i <= 10; i++)
        {
            mpDebugPortLevelSet(2, i * 10);
            mpTaskDelay((i + 2) / mpGetRtc() / 2);
        }
    }
}
```

5.5.9.4 Debug Port APIs

# mpDebugPortInit

Initializes the debug port for the port debug function. Before using the port, it is necessary to execute it once.

- **Syntax**

```
int mpDebugPortInit(
int portNo,
char* name
);
```

- **Argument**

[portNo]

Specify the port number from 0 to 4.

[name]

Specifies the name of the debug port. The port name which is specified in this step is displayed in the window of the port debug. For the port name, use a maximum of 32 characters.

- **Return value**

0: Normal end

-1: Error

# mpDebugPortHigh

Sets the output level of the debug port to High.

- **Syntax**

```
int mpDebugPortHigh(
int portNo,
);
```

- **Argument**

[portNo]

Specify the port number from 0 to 4.

- **Return value**

0: Normal end

-1: Error

# mpDebugPortLow

Sets the output level of the debug port to Low.

- **Syntax**
  int mpDebugPortLow(

  int portNo

  );

- **Argument**
  [portNo]

  Specify the port number from 0 to 4.

- **Return value**
  0: Normal end

  -1: Error

# mpDebugPortLevelSet

Sets the output level of the debug port to a specified value.

- **Syntax**
  int mpDebugPortLevelSet(

  int portNo,

  int level

  );

- **Argument**
  [portNo]

  Specify the port number from 0 to 4.

  [level]

  Specify the output level. The valid settable range is 0 to 100.

- **Return value**
  0: Normal end

  -1: Error

### 5.5.10 Copy Protection

This function allows the developer to provide copy protection to the application. The application will only run on an authorized controller.

Copy protection procedures are shown below.

1. **(Developer)** Write the protection activation process in the application source.
2. **(End User)** Acquire the serial file from the controller to use the application.
3. **(Developer)** Create a protection key from the acquired serial file.
4. **(End User)** Load the created protection key to the controller.

When protection of the application is enabled, the protection key is checked before starting the application. When the key is unauthorized or the key does not exist, the application does not start.

#### 5.5.10.1 Protection Activation

Add the following function (mpCopyProtectionActive()) in the source file of the application. If this function returns ON, the protection is enabled. If this function returns OFF, the protection is disabled.
If this function is not added to the source code, protection is disabled.

```
long mpCopyProtectionActive( void )
{
    return ( ON ); // ON: Activate protection, OFF: Inactivate protection
}
```

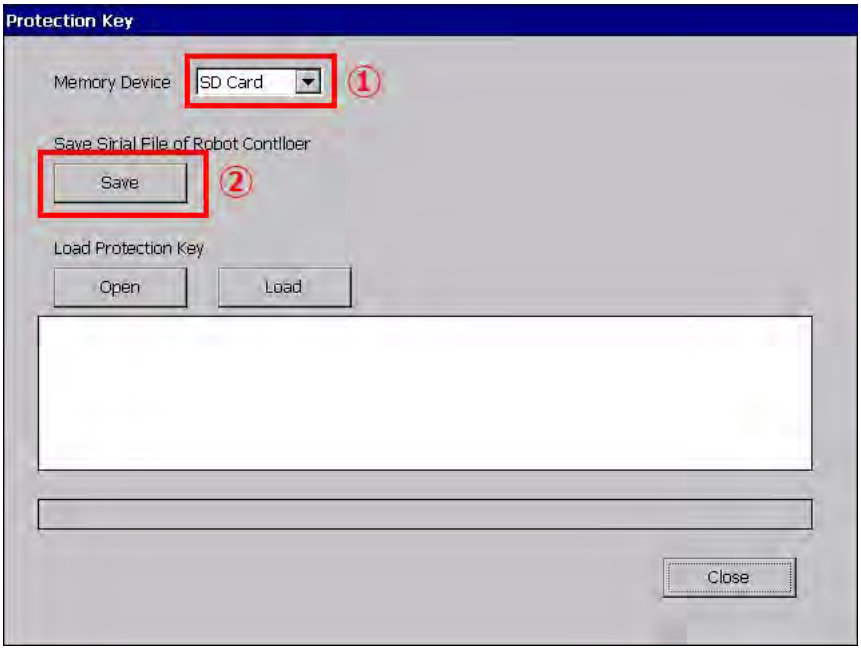
#### 5.5.10.2 Acquiring the Serial File (MotoPlus Monitor)

Acquires the serial file from the controller.

1. Select {Protection Key} in MotoPlus Monitor.
2. Select the Memory Device to save the serial file.
3. Press {Save}.

The serial file (MPSERIAL.DAT) is saved in the root folder of the selected memory device.

*Fig. 5-15: Serial File Acquisition Window*



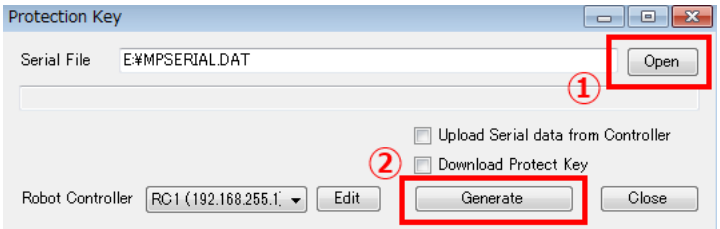
### 5.5.10.3 Generating the Protection Key

Generate the protection key from the acquired serial file.

1. Open the project of the application by using Visual Studio, and build once.
2. Select “M+ Protection Key” from the “Tools” menu.
3. Press {Open}, and select the acquired serial file (MPSERIAL.DAT).
4. Press {Generate}.

The key file (application name.KEY) is generated in the same folder as the OUT file of the application.

*Fig. 5-16: Protection Key Generation Window*




**NOTICE**

- The protection key cannot be generated if the project of the application does not exist.
- The protection key cannot be generated if the project of the application is not built.

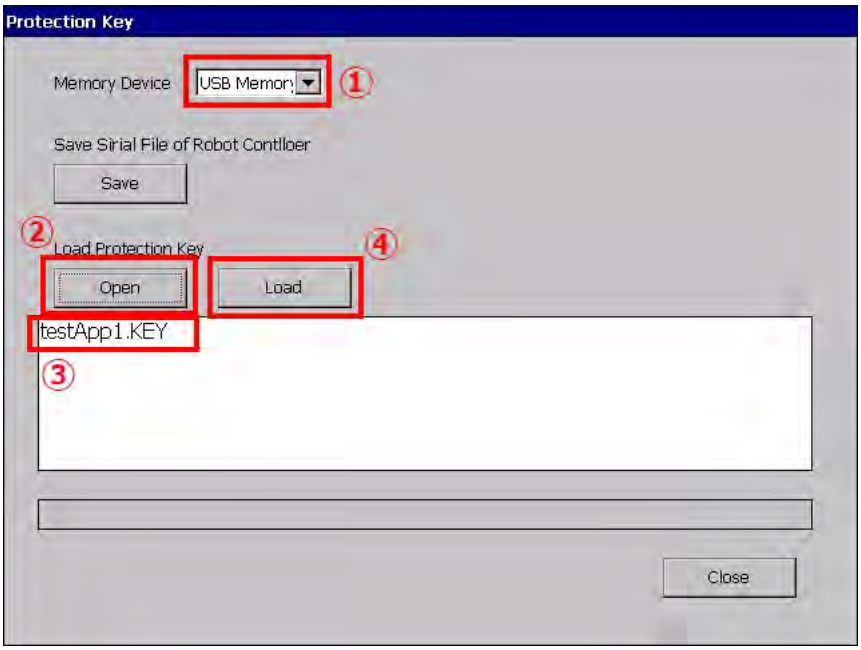
#### 5.5.10.4 Loading the Protection Key (MotoPlus Monitor)

Load the protection key to the controller.

1. Save the protection key file to the root folder of the memory device, and then insert the memory device into the programming pendant.
2. Select {Protection Key} in MotoPlus Monitor.
3. Select the memory device which the protection key is saved to.
4. Press {Open}.
5. The key file list on the memory device appears. Select the key file to load.
6. Press {Load}.

The protection key loads from the memory device to the controller. The application which is created in *section 5.5.10.1 "Copy Protection"* can be used. Copy protection procedures are complete.

*Fig. 5-17: Protection Key Download Window*



# 6 Alarm List

| Alarm Number | Alarm Name | Subcode | Meaning |
|---|---|---|---|
| 1020 | MOTOPLUS APPLICATION LOAD ERROR | 1 | Too many MotoPlus applications (.out files) are loaded in the controller. |
| | | 2 | The MotoPlus application is too large. (Available memory area is 2 Mbyte.) |
| | | 3 | Failed to open the directory. |
| | | 4 | Load failure (Failed to open the file.) |
| | | 5 | Load failure (Undefined symbol) (You declared an external function or variable that was never defined in the system) |
| | | 6 | Load failure (Others: application overloaded) (Verify that your .out file has been compiled for the correct robot controller) |
| | | 7 | Failed to initialize the API library. |
| | | 8 | No user root task. |
| | | 9 | Failed to create the user root task. |
| | | 10 | Failed to create RAM-Disk. |
| 4478 | MM TASK NO RESPONSE (MotoPlus) | | The process requested from the MotoPlus application to the MM task was not completed within the specified time. |
| 4479 | MOTOPLUS MM TASK WATCHDOG ERROR | | The man-machine task does not run for 3 seconds or more. |

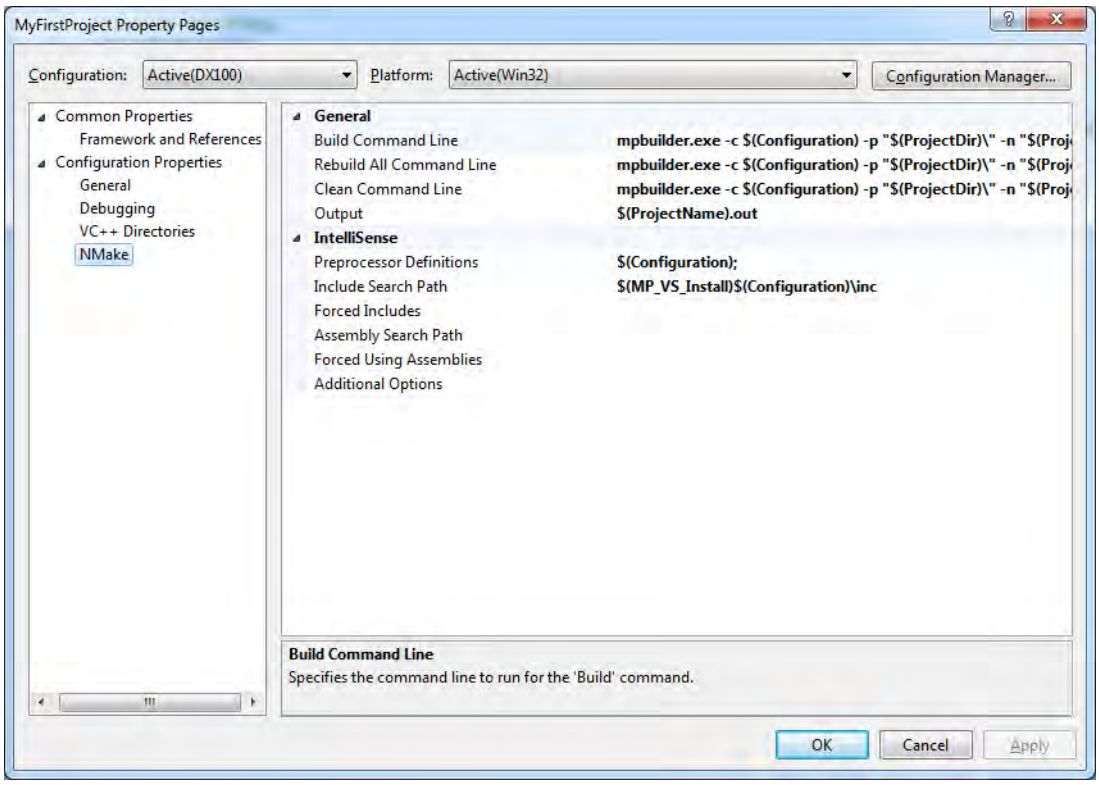
# 7 Advanced mpBuilder Usage

mpBuilder is the compile system utilized by MotoPlus for Visual Studio. This section describes the advanced usage for the build system. This should only be modified by experienced developers.

## 7.1 Visual Studio MakeFile Project

When select a MotoPlus Project from the New Project menu, the wizard is creating a Visual Studio MakeFile project. This project type in Visual Studio issues a command line call to a custom compiler.

The Build Command Line command can be viewed under {Project} > {Properties} > {Configuration properties} > {NMake}.




The default build command is:

```
mpbuilder.exe -c $(Configuration) -p "$(ProjectDir)\"
-n "$(ProjectName)" -b "$(OutDir)\" -o build
```

The default clean command is:

```
mpbuilder.exe -c $(Configuration) -p "$(ProjectDir)\"
-n "$(ProjectName)" -b "$(OutDir)\" -o clean
```

The fields $(…) are Microsoft macros for MSBuild. These are documented online on the MSDN website.

Using the $(…) syntax, you can also insert any system Environment Variables into the command line. Example: $(MP_VS_Install) evaluates to the MotoPlus installation directory.

The default commands do not use all of the available command-line arguments for mpBuilder. Here is a list of all available arguments:

- **-c**, --controller Required. Target YASKAWA robot controller platform. DX100, DX200, FS100 and YRC1000
- **-p**, --projectFolder Required. Root folder for the MotoPlus project.
- **-n**, --projectName Required. Name of your MotoPlus project.
- **-o**, --operation Required. Build operation to perform. 'build' = Build Project. 'clean' = Delete output files. 'rebuild' = Clean then build.
- **-b**, --binaryLocation Path to the output binary. If this is not specified, it will default to [ProjectDir]\output.
- **-i**, --includeDir This is the path to the standard include-directory for your MotoPlus API header files. If this is not specified, it will default to [InstallDir]\[Controller]\inc.
- **-e**, --encoding The character encoding of your source files. Example: “US-ASCII” or “UTF-8”. A list of supported encodings can be found in the Microsoft documentation for EncodingInfo.Name
- **-l**, --library 'Y' if you are building a MotoPlus Library file. 'N' if you are building a .out Executable module.
- **-d**, --dontParseErrors 'T' (true) if you want to see the unmodified error/warning strings returned from the compiler. Otherwise, they will be parsed and formated to be recognized by Visual Studio.
- **-v**, --visualStudioVersion Must be specified for VS2008 (v=9). Ignored for Visual Studio 2010 - 2013.

## 7.2 Compiler Arguments

The specific compiler will vary based on which robot controller is selected. The proper one is selected based on the arguments specified for mpBuilder. Regardless of which controller is selected, the final compiler will be part of the GCC build system.

The arguments for the GCC build system are specified in the .mps files, which are automatically added to your MotoPlus for Visual Studio project.

| NOTICE |
| --- |
| Please see the GCC documentation for all available arguments to the build system. |

### 7.2.1 CompilerArguments.mps

This file is used when compiling source files into binary objects. This can be used to control many variables in the build system.

- Project-wide DEFINE statements
- Enable/disable particular errors and warnings
- Compiler code optimization (not recommended)
- Directories to search for include-files for the project

There are macros which can be placed in the .mps file which will be recognized by mpBuilder and resolved prior to calling the compiler.

- ~ProjectDir~ : This will resolved to the temporary directory where the source files are copied and converted to ASCII encoding at build time.
- ~IncludeDir~ : This is the standard include directory to search for MotoPlus API header files. If the -i argument is specified to mpBuilder.exe, then this macro will resolve to the value of the -i argument. Otherwise, it will default to [MP_VS_Install]\[Controller]\inc
- ~FilePath~ : This will resolve to the full path of the source file being compiled. Please note that it will point to the ASCII copy of the source file, located in the temp directory.
- ~OutputPath~ : This will resolve to the full path of the compiled object file which is the output from the compiler.

### 7.2.2 LinkerArguments.mps

This file is used when linking the compiled object files into a single binary executable/library. There are macros which can be placed in this .mps file which will be recognized by mpBuilder and resolved prior to calling the compiler.

- ~FileList~ : This is the list of compiled object files which will be combined into the final output file.
- ~OutputPath~ : This is the full path to the final output file.

# Appendix A

**NOTICE**

The following compatibility issues are not applicable to MotoPlus SDK v1.1.0 or higher. The fixes below are only required for older versions.

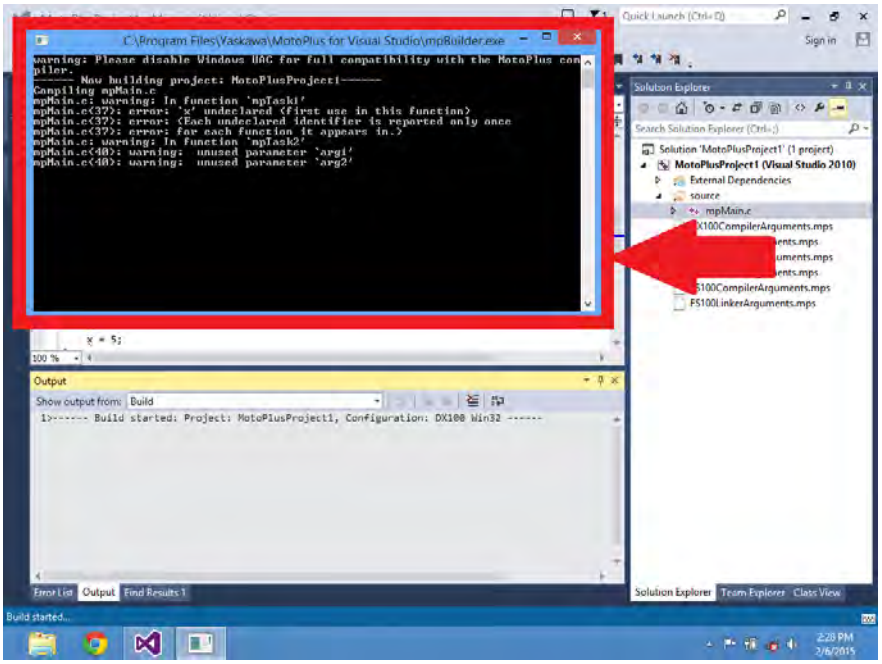
## A.1 Compatibility with Windows 8/10 (MotoPlus SDK v1.0.7 and Lower)

The MotoPlus compiler is compatible with Windows 8. However, additional steps must be performed to make this work properly.

### A.1.1 Symptom of Problem:

If a console window appears when attempting to build your MotoPlus application, then the compiler will not properly display Errors/Warnings inside of Visual Studio.

*Fig. A-1: Compiler Not Displaying Errors/Warning Inside of Visual Studio*



- **Solution:**
  You must explicitly run Visual Studio as Administrator each time you launch the application. Please note that simply having an Administrator user account is not sufficient. When launching Visual Studio, you must right-click and select {Run as Administrator}.

| NOTICE |
| --- |
| You must perform this every time you want to open Visual Studio. |

*Fig. A-2: From Metro Interface*

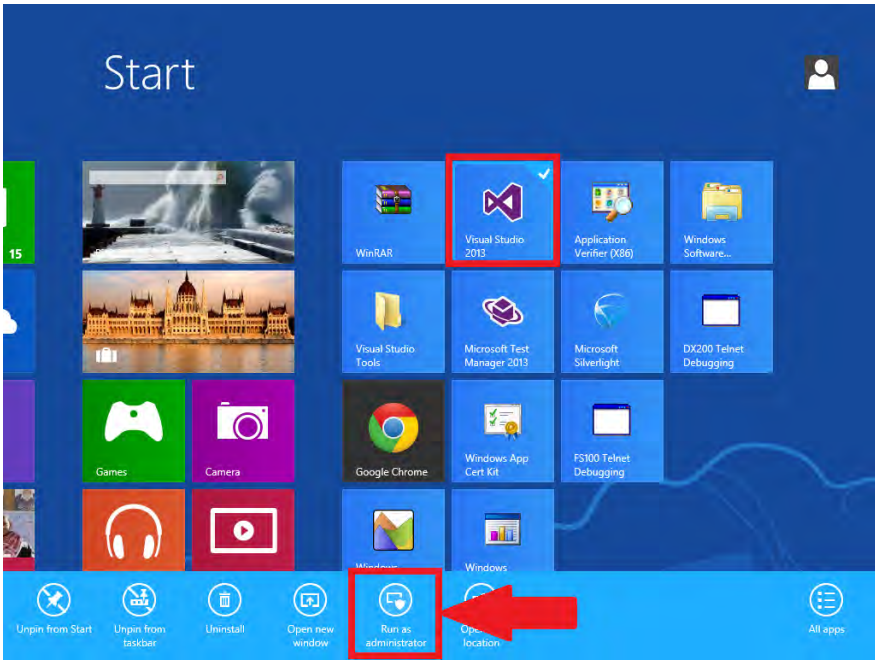
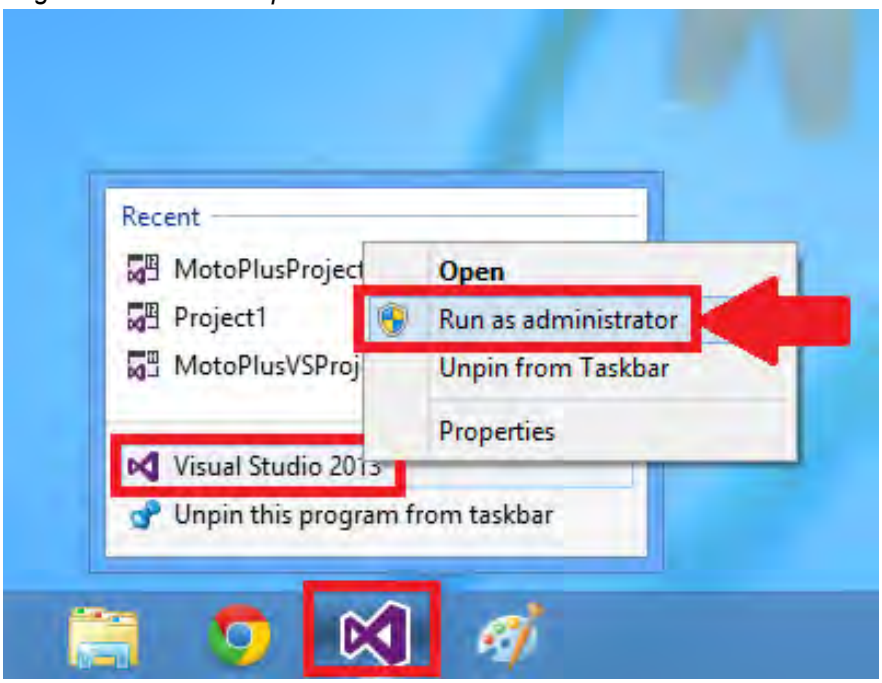


*Fig. A-3: From Desktop Interface*



# Appendix B

## B.1 Compatibility with Visual Studio 2015

| CAUTION |
| --- |
| • Update Visual Studio 2015 to “update 2” or greater.<br>Not updating Visual Studio to “update 2” or greater will prevent compiling MotoPlus apps. |

MotoPlus SDK for Visual Studio is compatible with Visual Studio 2015.

### B.1.1 Symptom of Problem:

If you see the error message “The filename, directory name, or volume label syntax is incorrect,” then you must update Visual Studio.

# MOTOPLUS SDK
# FOR VISUAL STUDIO
# USER'S MANUAL

**FOR: DX100, DX200, FS100, YRC1000, AND YRC1000micro**
**YASKAWA ELECTRIC CORPORATION**

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